Mapping Cancer Therapy Mechanisms, Evidence Gaps, and Resistant-State Hypotheses: A Cross-Literature Analysis

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Abstract

Cancer-therapy literatures often mature in parallel without being compared on shared axes such as evidence depth, resistant-state biology, pathway dependencies, or delivery constraints. Here we present a cross-literature analysis of 4,830 full-text cancer research articles (803 journals, 19 mechanisms, 22 cancer types, 2001-2026), alongside a separate archive of 5,584 abstract-only records, to map where research is concentrated, where apparent gaps are likely to be artifacts, and which mechanistic ideas deserve broader comparison. We measure the quality of our own evidence tagger (46% exact-label accuracy, 96% binary precision, 55% recall on a 100-article gold set), derive tissue-of-origin and weighted-evidence layers to sharpen mechanism comparisons, and introduce a diagnostic-to-therapy matching framework covering six translational chains. Monte Carlo and spatial simulations explore ferroptosis biochemistry across cell phenotypes, tissue depths, drug penetration gradients, drug combinations, and oxygen-mediated tumor microenvironment effects. Key simulation findings: (1) RSL3 + FSP1 inhibitor produces 1.99× Bliss synergy through dual-pathway depletion, (2) RSL3 efficacy collapses from 3.7% to 0.1% under tumor hypoxia while SDT maintains 87.8% kill because exogenous ROS bypasses the mitochondrial bottleneck, and (3) tissue-specific drug penetration reduces RSL3 kill from 40% (2D) to 1.8% behind the blood-brain barrier. Within this broader landscape, one specific hypothesis survives as worth testing rather than assuming: physical ROS-generating modalities, especially sonodynamic therapy (SDT), may maintain efficacy in conditions — hypoxia, deep tissue, limited drug penetration — where pharmacologic ferroptosis inducers fail. We frame this as a conditional implication of the wider analysis, not the article's starting assumption. The result is a more cautious but more useful conclusion: the important question is not simply which modality looks interesting, but which interventions remain plausible after accounting for resistance state, tissue access, tumor microenvironment, evidence maturity, and known corpus artifacts. All simulation code (MIT-licensed Rust library with Python bindings), analysis pipelines, and data are openly available for replication and extension.

Keywords: cross-literature analysis, resistant states, evidence gaps, ferroptosis, photodynamic therapy, sonodynamic therapy, radioligand therapy, immunotherapy, pathway targets, cancer therapy

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Part I

Why This Exists

Chapter 1

The Problem

1.1 The Scale of Loss

Cancer accounted for nearly 10 million deaths worldwide in 2022, making it one of the leading causes of death on Earth [News: WHO, "Cancer Fact Sheet", who.int, 2024. Verified: IARC GCO]. Lung cancer alone produced 2.5 million new cases that year. Behind each of those numbers is a person, a family, a dinner table with an empty chair. The statistics are so large that they lose their human weight — but the weight is there.

This is not a new crisis. Governments, pharmaceutical companies, academic institutions, and advocacy organizations have spent decades and hundreds of billions of dollars fighting cancer. And in some areas, the investment has paid off remarkably well. Five-year survival rates for childhood acute lymphoblastic leukemia — once a near-certain death sentence — now exceed 90% (NCI SEER, 2014-2020). In the 1960s, a child diagnosed with ALL faced roughly a 10% chance of surviving five years. Today, through a combination of multi-agent chemotherapy, risk-stratified treatment protocols, and improvements in supportive care, that number has climbed above 90%. Female breast cancer, caught at the localized stage, has a five-year survival rate of nearly 100% (NCI SEER, 2015-2021). Overall, breast cancer survival stands at 91.7%, reflecting decades of investment in screening, surgical technique, and targeted therapies such as tamoxifen and trastuzumab. Testicular cancer is among the most curable solid tumors, with overall cure rates exceeding 95%.

These are genuine triumphs. They represent what cancer medicine can accomplish when early detection is possible, when effective therapies exist, and when the biology of the disease cooperates.

But the picture is radically uneven. Pancreatic cancer has an overall five-year survival rate of 13.3% (NCI SEER, 2015-2021). More than half of patients are diagnosed only after the disease has spread to distant sites, where survival drops to 3.2%. Even among the 15% of patients whose cancer is detected at the localized stage, the five-year survival

rate is only 43.6% — far below the near-100% rates seen for localized breast cancer. Pancreatic ductal adenocarcinoma is lethal in approximately 88% of patients LA et al. [2023], a statistic that has improved only marginally over the past several decades.

Glioblastoma, the most common malignant primary brain tumor, has a median survival of less than two years even with maximal treatment including surgery, radiation, and chemotherapy AC et al. [2020]. The addition of tumor-treating fields — a novel device-based approach — extended median survival to approximately 21 months compared to 16 months with chemotherapy alone R et al. [2017], a meaningful improvement but still measured in months, not years.

Lung cancer, despite the revolution in immunotherapy, still carries a 28.1% five-year survival rate overall (NCI SEER, 2015-2021). For patients diagnosed with distant-stage disease — which represents the majority of cases for several lung cancer subtypes — that number falls to 9.7%.

The contrast is stark. For cancers where early detection works and biology cooperates, survival rates approach 100%. For cancers that are diagnosed late, grow in hard-to-reach locations, and harbor cells that resist treatment, survival rates remain in the single or low double digits despite decades of effort.

The question is not whether cancer research is making progress. It is. The question is why progress is so uneven — and what keeps certain cancers stubbornly resistant to everything we throw at them.

1.2 Where Treatments Stall

The cancers with the worst outcomes share a pattern. They tend to be diagnosed late, when the disease has already spread. They tend to involve tissues that are hard to reach surgically or with drugs. And critically, they tend to harbor cells that survive initial treatment and come back stronger.

Consider the trajectory of a patient with pancreatic ductal adenocarcinoma. The disease is lethal in approximately 88% of patients LA et al. [2023]. Surgery is possible in only a minority of cases, because most tumors are locally advanced or metastatic at diagnosis. Chemotherapy can shrink tumors and extend survival by months, sometimes years. But in the vast majority of cases, the disease returns. Not because the chemotherapy did nothing — it often does a great deal — but because a small population of cells survived, adapted, and eventually re-established the tumor.

The same pattern appears in glioblastoma, where surgical resection followed by radiation and temozolomide produces a median progression-free survival of approximately 7 months before the tumor recurs. It appears in metastatic melanoma, where immunotherapy produces dramatic responses in some patients but leaves others with progressive disease driven by resistant clones. It appears in ovarian cancer, where platinum-based

chemotherapy often produces an initial complete response, only for the disease to recur in a platinum-resistant form.

This pattern — initial response followed by relapse driven by surviving cells — is the central obstacle to durable remission. It is the reason why a 99% tumor reduction can still end in death. The cancer doesn't need to outsmart the treatment at every turn. It only needs to survive it once.

Understanding why those cells survive, what makes them different from the cells that responded, and whether their survival creates new vulnerabilities that could be exploited before they recover — these are among the most important questions in modern oncology.

1.3 Why Treatments Fail

Three biological realities explain why cancer is so difficult to eradicate, even when initial treatment works well.

Tumors are populations, not clones. A single tumor can contain billions of cells, and not all of them are the same. They carry different mutations, express different proteins, and rely on different metabolic strategies. When a drug targets one characteristic — say, a specific mutation — it kills the cells that depend on that characteristic. But the cells that don't depend on it survive. The tumor doesn't fight back in any conscious sense. It is simply a diverse population being filtered by selective pressure.

This is the same principle that drives antibiotic resistance in bacteria. When you expose a bacterial population to an antibiotic, the resistant survivors repopulate the colony. The antibiotic didn't create resistance; it selected for it from the existing diversity. Cancer operates on the same selection principle, though the mechanisms of diversification differ — bacteria can share resistance genes between cells, while tumor cell diversity arises primarily from mutation and epigenetic changes during cell division.

Drug resistance is an almost inevitable consequence of treatment. The cells that survive a first round of chemotherapy are, by definition, the ones that had some trait allowing them to survive. When those cells divide, they produce progeny that share that trait. Over time, the population shifts from drug-sensitive to drug-resistant. This is why many targeted therapies that produce dramatic initial responses — shrinking tumors by 90% or more — eventually fail. The responding cells are eliminated; the resistant cells remain and expand into the space left behind.

Resistance can emerge through many routes. Some cells physically pump drugs out before they can take effect, using molecular transporters that act like bilge pumps ejecting water from a leaking ship. Others alter the drug's molecular target so the drug can no longer bind to it — like changing a lock so the key no longer fits. Still others activate alternative survival pathways that bypass the one being blocked, maintaining cell survival through a different route even when the original pathway is shut down. And some cells

simply stop dividing, entering a dormant state where they consume less energy, present fewer targets, and wait out the treatment like a bear in hibernation.

The details vary by cancer type and treatment, but the principle is consistent: **selective pressure drives adaptation.** This is not a bug in cancer biology. It is a fundamental feature of how evolution works at the cellular level, and it operates on timescales of weeks to months — fast enough to undermine even initially successful therapies.

Killing most of a tumor is not the same as curing the disease. This is perhaps the hardest reality for patients and families to understand. Imaging shows the tumor has shrunk. Blood markers have dropped. The clinical response looks excellent. But if even a small population of altered cells persists — hidden in tissue niches, metabolically dormant, invisible to standard detection — those cells can eventually reawaken and seed a recurrence that is harder to treat than the original.

This is not a failure of medicine so much as a reflection of how evolution works at the cellular level. Treatment changes the selective landscape. The cells that emerge from that landscape are the ones that were least affected by the treatment. And the next treatment must contend with a population that has already been filtered for survival.

1.4 The Residual Disease Problem

The cells that survive treatment are not merely leftovers. Increasingly, research shows that treatment itself can push cancer cells into altered states — sometimes called drug-tolerant persister states — where their biology is fundamentally different from the original tumor M et al. [2026].

These residual cells may shift their metabolism, relying on different energy sources than the cells that responded to treatment. Where the original tumor might have relied heavily on rapid glucose consumption — the so-called Warburg effect — a preference for fermenting glucose even when oxygen is available — that has defined cancer metabolism for over a century — the surviving cells may switch to mitochondrial respiration, fatty acid oxidation, or other metabolic strategies that are less affected by the treatment that just killed their neighbors.

They may enter a state of suspended proliferation, dividing slowly or not at all. This makes them less visible to drugs that target rapidly dividing cells and less detectable by standard imaging methods that rely on metabolic activity. They may also upregulate protective mechanisms — molecular defenses that scavenge the toxic byproducts of cellular stress and repair damage faster than it accumulates.

And critically, these altered states may be temporary. Given time — days, weeks, perhaps months — the cells can revert, resume growing, and establish a recurrence that carries the resistance lessons learned from the first treatment. The clock is ticking in both directions: the patient needs the residual cells eliminated before they recover, and

the residual cells need time to restore the defenses that will make them resistant to the next round of treatment.

This creates both a challenge and an opportunity. The challenge is obvious: residual cells are hard to find, hard to target, and biologically different from the cells we know how to kill. Standard approaches that worked against the original tumor may not work against the survivors because the survivors are not the same cells.

The opportunity is less obvious but potentially more important: if the altered state that makes these cells resistant to one type of attack also makes them vulnerable to a different type of attack, there may be a window — possibly brief — during which they could be eliminated before they recover. Whether such a window reliably exists in human tumors, how long it lasts, and whether any therapeutic approach can exploit it in time are open questions, not established facts.

This concept — that resistance to one insult might create sensitivity to another — is not new in biology. It has been observed in antibiotic combination therapy, where bacteria resistant to one antibiotic sometimes become hypersensitive to another. It appears in the concept of collateral sensitivity, where the evolutionary trade-offs required for resistance to one drug create dependencies that a second drug can exploit. And it is the foundation of synthetic lethality approaches in cancer, where two genetic events that are individually survivable become lethal in combination.

What is less clear is which specific vulnerabilities residual cancer cells develop, how long those vulnerabilities persist, which therapeutic approaches could realistically exploit them, and whether the exploitation window is long enough to matter clinically. A vulnerability that lasts 48 hours is biologically real but may be clinically useless if the treatment can't be delivered in time. A vulnerability that lasts three months is clinically actionable — but does it actually exist?

These are the questions that motivated this project. Not the claim that one particular approach is the answer, but the recognition that answering these questions requires comparing across therapeutic modalities and biological mechanisms in a way that the current literature — organized by individual modalities and studied within individual research communities — does not easily support.

1.5 The Current Landscape

The field is not short of promising approaches.

Immune checkpoint inhibitors have transformed outcomes for some cancers. The immune system has an inherent ability to recognize and attack cancer cells, but tumors often deploy molecular signals that tell immune cells to stand down. Checkpoint inhibitors block those signals, releasing the brakes on the immune response. Drugs such as pembrolizumab and nivolumab have produced durable responses in melanoma, non-small-cell

lung cancer, renal cell carcinoma, and several other tumor types — in some cases, converting previously terminal diagnoses into long-term survivorship. But the response is far from universal. Many patients do not respond at all. Among those who do, resistance can develop over time. And the tumor microenvironment — the complex ecosystem of immune cells, blood vessels, structural proteins, and signaling molecules that surrounds the tumor — plays a decisive role in determining whether immunotherapy succeeds or fails. "Cold" tumors that exclude immune cells remain largely impervious to checkpoint blockade.

Engineered cell therapies, particularly chimeric antigen receptor T cells (CAR-T), represent a different strategy: rather than releasing the brakes on existing immune cells, they manufacture new immune cells specifically designed to attack cancer. CAR-T therapy has been transformative in certain blood cancers. Patients with previously untreatable B-cell leukemias and lymphomas have achieved complete remissions that would have been unthinkable a decade ago. But extending this success to solid tumors has proven far more difficult. Solid tumors create hostile environments that suppress immune cells attempting to infiltrate them, surround themselves with physical barriers of connective tissue and extracellular matrix (the structural scaffolding between cells), and present a moving target as tumor cell populations evolve and change their surface markers.

Targeted therapies — drugs designed to exploit specific molecular features of a patient's cancer — represent the precision medicine ideal. When a tumor depends on a single mutated protein for its growth, a drug that blocks that protein can produce rapid, dramatic shrinkage. But resistance is the rule rather than the exception. Within months, cancer cells find alternative molecular routes to the same outcome, bypass the blocked pathway entirely, or mutate the drug target so it is no longer recognizable. The targeted therapy literature is filled with stories of dramatic initial responses followed by inevitable relapse.

Newer modalities are advancing through clinical trials and early adoption. Radioligand therapy attaches radioactive atoms to molecules that seek out specific receptors on cancer cells, delivering targeted radiation directly to tumors while sparing surrounding tissue. Bispecific antibodies are engineered proteins that physically bridge immune cells and cancer cells, forcing them into contact. mRNA cancer vaccines train the immune system to recognize tumor-specific mutations unique to each patient's cancer. Antibody-drug conjugates deliver toxic payloads directly to cancer cells using antibody guidance, like molecular guided missiles. Each of these approaches addresses a real limitation of earlier strategies, and several are producing encouraging results in clinical trials.

The challenge is not that we lack ideas. It is that each approach is typically developed, studied, and promoted within its own research community. The immunotherapy literature compares checkpoint inhibitors to each other. The cell therapy literature compares CAR-

T designs. The targeted therapy literature compares small molecules against specific mutations. But the questions that matter most to patients — which approaches work under which conditions, for which types of resistance, in which tissue contexts — require comparisons that cut across these boundaries.

That cross-boundary comparison is what this project attempts.

Chapter 2

What We Set Out To Do

2.1 The Open-Science Bet

This project started from a simple conviction: breakthroughs against diseases that destroy lives should be a human right, not a revenue stream. That conviction shapes every decision in this repository, from the MIT license on all code to the publicly available data and the openly documented analysis pipeline.

Open science is not a marketing strategy. It is a deliberate bet that the work has more value in the hands of the global research community than behind any paywall or patent wall. Every script, every simulation, every analysis document, and every draft of this manuscript is publicly accessible. Anyone can re-run the analysis from scratch, challenge the conclusions, extend the simulations, or take an idea in a direction we never considered. If someone takes something from this work, validates it in a lab, and it helps a patient — that is the point.

The trade-off is real. Open-sourcing everything means forgoing patent protection, proprietary advantage, and the kind of controlled dissemination that academic publishing traditionally provides. We accept that trade-off because the work has more value in the open than behind a paywall. If someone takes an idea from this work, validates it in a lab, and it helps a patient — that matters more than who gets credit.

Concretely, this means: all code is MIT-licensed. The ferroptosis-core simulation engine is published as a standalone Rust library with Python bindings, so any researcher can use it independently of this manuscript. The corpus analysis pipeline is fully documented and re-runnable. The repository welcomes contributions — open an issue, submit a pull request, fork the entire project and take it somewhere better. Nothing is hidden, nothing requires special access, and nothing requires our permission.

2.2 Two Computational Lenses

This project uses two complementary computational approaches, neither of which replaces laboratory experiments but both of which can help focus them.

Cross-literature analysis. We used software to systematically read and categorize 4,830 full-text cancer research articles from 803 journals, spanning 19 therapeutic mechanisms and 22 cancer types. A separate archive of 5,584 abstract-only records provides additional coverage. The software tags each article by mechanism, cancer type, evidence level, tissue of origin, and other categories, then maps the relationships between them.

This reveals patterns that individual researchers are unlikely to see, because most researchers read within their field. An immunotherapy specialist reads immunotherapy papers. A nanomedicine researcher reads nanoparticle papers. A ferroptosis biologist reads ferroptosis papers. Each sees their own literature clearly but may not know what adjacent fields have found — or failed to find — about the same biological questions. Are the mechanisms that make cancer cells resistant to one type of treatment the same mechanisms that another field has already learned to exploit? Are two different research communities studying the same biological process under different names? Is a "gap" in one field's literature actually well-covered in another? These are the kinds of questions that emerge when you read across fields rather than within them.

The cross-literature analysis maps the full landscape so that comparisons across boundaries become visible. It does not replace expert judgment. It maps terrain; experts navigate it. And it has known limitations: the evidence tagger that classifies articles has 96% precision but only 55% recall, meaning what it tags is almost always real, but it misses a substantial share of valid evidence. Absence in our data is not absence in the field. Every gap claim in this book should be read as "not detected in our keyword-based analysis of 4,830 articles" rather than "does not exist."

Monte Carlo biochemical simulation. We built a computational model of a specific cell-death pathway — ferroptosis, a type of cell death driven by iron-catalyzed damage to the lipid membranes that surround cells — and ran it millions of times with random variation to understand which conditions make cells more or less vulnerable.

Monte Carlo simulation is named after the casino in Monaco, but the output is statistical knowledge, not luck. By running the same model thousands or millions of times with slightly different starting conditions, we generate not a single answer but a distribution of possible outcomes. This tells us which results are robust (they happen every time regardless of variation) and which are fragile (they depend on specific parameter values being exactly right). When a simulation result holds across a wide range of random variation, it suggests a pattern worth investigating in the laboratory. When it doesn't, it flags an assumption that needs experimental verification.

The simulations are hypothesis-generating tools. They identify which biological ques-

tions are worth asking and which parameter combinations lead to interesting outcomes. They do not prove that any particular treatment works. Computation without experimental validation is speculation. But experiment without computational guidance is inefficient when there are more hypotheses to test than any laboratory can pursue.

2.3 What We Can and Cannot Contribute

The value of this work lies in the questions it raises and the comparisons it surfaces. Every finding in this book should be read as: "Under these assumptions, the model suggests X. Here is what would need to be true for X to hold in reality, and here is what would falsify it." That framing — hypothesis with falsification criteria — is the contribution.

What constrains it: we are not clinicians, and nothing here is medical advice or treatment guidance. We are not experimentalists — our simulations use estimated parameters and produce directional insights, not quantitative predictions. A simulation showing that drug efficacy drops under low-oxygen conditions identifies a mechanism worth testing; it does not predict what will happen in a specific patient.

Our corpus is 4,830 full-text articles — a meaningful sample but a small fraction of PubMed's millions of records. The archive was constructed from keyword queries targeting our 19 mechanism categories, which means articles using non-standard terminology, articles in non-English languages, and articles outside our search scope are not included. We also maintain 5,584 abstract-only records that extend coverage without full-text depth. When we report that a combination has not been studied, we mean it was not detected in our local analysis — not that it does not exist. Some landmark papers are known to be missing.

Our evidence tagger has 96% precision but only 55% recall: what we tag is almost always real, but we miss nearly half of valid evidence. We have documented these gaps throughout the book — not to excuse them, but because transparency about what we don't know is as important as reporting what we found.

2.4 How To Read This Book

This book is written for multiple audiences. Not every chapter is equally relevant to every reader.

Researchers in oncology may want to skip Part I and start at Chapter 3, which maps the literature landscape across all 19 mechanism categories. Chapter 7 presents four tumor microenvironment features — hypoxia, immune coupling, stromal shielding, and acidic pH — that the simulations suggest selectively disadvantage drug-based approaches while leaving physical modalities relatively unaffected. Chapter 9 translates simulation

findings into specific experimental proposals with expected outcomes and falsification criteria.

Graduate students will benefit from reading cover to cover. Part I provides the human and scientific context. Part II demonstrates how a cross-literature analysis works in practice — including where it fails and how to interpret its limitations. Part III shows computational biology applied to real research questions, with every model assumption documented and every result accompanied by caveats. Part IV describes where the research goes next and how to contribute.

Clinicians may find Chapter 1 familiar territory — the problem of treatment resistance and residual disease is what you contend with daily. Chapter 7 quantifies microenvironment mechanisms that affect drug delivery and efficacy in solid tumors. Chapter 9 describes the experimental directions that could eventually inform clinical practice.

Engineers and software developers should look at Chapter 5, which describes the simulation architecture, and Chapter 11, which explains how to extend the computational work. The ferroptosis-core library is a standalone MIT-licensed Rust crate with Python bindings via PyO3. Appendix B provides a step-by-step reproduction guide for every analysis and simulation in the book.

Part I has motivated the problem and described the project. It has not made therapeutic recommendations or presented results. For the evidence, turn to Part II.

Part II

What We Found

Chapter 3

The Literature Landscape

3.1 Introduction

Cancer research is rich in promising mechanisms but poor at comparing them on a common frame. Fields grow around modalities, delivery platforms, or molecular targets, yet the real translational questions often cut across those boundaries: which interventions have credible human evidence, which are trapped in preclinical loops, which resistant states are actually being targeted, and which conclusions are artifacts of search design rather than biology. This problem becomes acute when one literature becomes rhetorically compelling before it has been properly compared against adjacent approaches.

Drug-tolerant persister cells illustrate the problem clearly. These residual populations are a central obstacle to durable remission, and recent work shows that some persister states are sensitized to ferroptosis, an iron-dependent cell-death pathway driven by lethal lipid peroxidation M et al. [2026]. That finding immediately raises a therapeutic question, but not just the obvious one. The field has focused largely on pharmacologic ferroptosis inducers such as RSL3, erastin, and GPX4-directed strategies G et al. [2022]. A broader question is whether resistant-state vulnerabilities should sometimes be matched to different delivery logics entirely, including spatially selective physical modalities, immune-engaging strategies, or pathway-directed approaches that sit outside the original literature silo.

This article therefore takes a broader starting point than our earlier framing. We first map the full-text corpus across 19 mechanism tags and 22 cancer types, assess evidence maturity, and highlight where gap claims are unstable because of taxonomy, missing full text, or incomplete evidence coverage. We then use that broader landscape to evaluate one narrower implication: whether physical ROS-generating modalities such as photodynamic therapy (PDT), sonodynamic therapy (SDT), and more speculatively cold atmospheric plasma (CAP) should be treated as candidate tools for locally targetable, ferroptosis-prone residual states rather than as generic alternatives to drugs. This reframing matters

because PDT/SDT are not the only adjacent directions that emerge when the repo is examined without modality loyalty. Radioligand therapy, phagocytosis checkpoints, targeted protein degradation, stromal shielding, senescence programs, and lipid-remodeling defenses all compete for attention once the analysis is widened.

We present a systematic analysis of 4,830 full-text cancer research articles spanning 19 therapeutic mechanisms and 22 cancer types (803 journals, 2001-2026), with a separate archive of 5,584 abstract-only records, together with simulation work on ferroptosis sensitivity across cell phenotypes. The aim is not to prove that one modality is the answer. It is to identify which hypotheses remain plausible after accounting for evidence depth, known corpus artifacts, and the biology of residual disease.

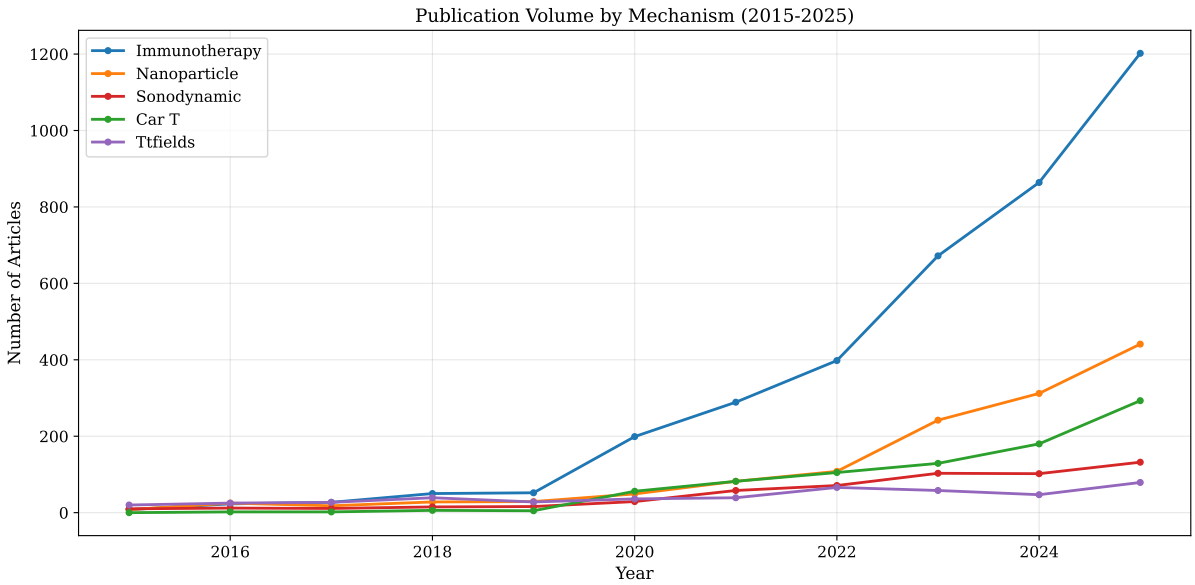

Figure 3.1: Publication volume 2015–2025.

3.2 Search Strategy and Corpus Construction

We conducted a systematic literature search of PubMed using the NCBI E-utilities API, covering publications indexed through March 2026. The search strategy employed mechanism-specific query sets using established MeSH terms and free-text synonyms: for example, "tumor treating fields" OR "TTFields" OR "Optune" for TTFields; "chimeric antigen receptor" OR "CAR-T" OR "CAR T cell" for CAR-T therapy; "sonodynamic therapy" OR "sonosensitizer" for sonodynamic approaches. During revision, we also identified additional underrepresented areas for future corpus expansion, including CAP, radioligand therapy/theranostics, targeted protein degradation, phagocytosis checkpoints, autophagy, senescence/SASP, stromal biology, cuproptosis, and disulfidptosis. The quantitative results reported in this manuscript still refer to the frozen 19-mechanism corpus

analyzed here; the expanded taxonomy is a next-pass scaffold rather than a retroactive redefinition of the present results. Results were enriched with metadata from OpenAlex (open access status, citation counts, topic classification) and PubMed Central or publisher-hosted OA URLs (full-text retrieval for open access articles).

3.3 Inclusion and Exclusion Criteria

Articles were included if they were returned by the PubMed search queries and addressed at least one of the 19 therapeutic mechanisms in the context of cancer treatment or cancer biology. No additional manual inclusion or exclusion filtering was applied beyond the search query constraints. The resulting archive therefore includes original research, clinical trial reports, systematic reviews, narrative reviews, and some case series. For quantitative analyses in this manuscript, we indexed only the 4,830 records for which local full text was available; the remaining 5,584 metadata/abstract-only records were retained in a separate archive and not used for the full-text-derived corpus statistics reported below. This distinction is important: article counts in the Results section refer to the full-text corpus unless explicitly stated otherwise.

3.4 Categorization Framework

Each article was tagged along three axes:

Therapeutic mechanism(s): Articles were assigned one or more of 19 tags representing distinct therapeutic modalities. We use "mechanism" broadly to encompass molecular targets (e.g., synthetic lethality), drug classes (e.g., ADCs), delivery platforms (e.g., nanoparticles), device-based interventions (e.g., TTFields), and biological strategies (e.g., microbiome modulation). This heterogeneous taxonomy reflects the diversity of the contemporary oncology toolkit; we prioritized practical distinctness over strict ontological consistency. The 19 tags are: (1) antibody-drug conjugate (ADC), (2) bioelectric modulation, (3) bispecific antibody, (4) CAR-T cell therapy, (5) CRISPR gene editing, (6) electrochemical therapy (including irreversible electroporation), (7) electrolysis, (8) epigenetic therapy, (9) frequency therapy (including pulsed electromagnetic fields), (10) high-intensity focused ultrasound (HIFU), (11) checkpoint immunotherapy, (12) mRNA cancer vaccine, (13) metabolic targeting, (14) microbiome modulation, (15) nanoparticle delivery, (16) oncolytic virotherapy, (17) sonodynamic therapy, (18) synthetic lethality, and (19) tumor-treating fields (TTFields). Articles addressing multiple mechanisms received multiple tags, enabling convergence analysis.

Cancer type(s): Articles were assigned to one or more of 22 cancer types: bladder, breast, cervical, colorectal, esophageal, gastric, glioblastoma, head-and-neck, kidney,

leukemia, liver, lung, lymphoma, melanoma, mesothelioma, myeloma, neuroblastoma, ovarian, pancreatic, prostate, sarcoma, and thyroid.

Evidence level: Original-research-like articles were classified into one of seven tiers: (1) Phase III clinical trial or randomized controlled trial, (2) Phase II clinical trial, (3) Phase I clinical trial, (4) non-phase clinical / patient-series evidence ('clinical-other'), (5) preclinical in vivo, (6) preclinical in vitro, or (7) theoretical/computational. Review-like articles (reviews, systematic reviews, meta-analyses, evidence maps) and protocols were intentionally left unclassified rather than being forced into an evidence tier. Classification was based on keyword detection plus publication-type metadata for the highest level of evidence reported in the article text. The 'clinical-other' bucket is deliberately conservative: it indicates patient-level evidence exists, but should not be interpreted as equivalent to phase-labeled trial maturity.

3.5 Analytical Approach

Four primary analyses were conducted:

Mechanism-cancer matrix: A 19 x 22 cross-tabulation of article counts by mechanism and cancer type, identifying concentration patterns and gaps.

Convergence map: Identification and quantification of all mechanism-pair co-occurrences across the corpus, revealing which multi-mechanism combinations are most actively researched.

Evidence-tier analysis: Assessment of the highest clinical evidence level achieved for each mechanism, identifying translational bottlenecks.

Gap analysis: Identification of mechanism-cancer and mechanism-mechanism pairs with zero or near-zero co-occurrence in our corpus, noting that some gaps reflect search query design rather than genuine research absence.

Impact metrics (citation counts, iCite Relative Citation Ratio) were used to identify landmark publications and assess field-normalized influence. Publication volume was tracked annually to identify growth trajectories.

3.6 Limitations

Several limitations should be acknowledged. First, all tagging was performed by automated keyword matching against curated dictionaries (no manual article-level review), introducing potential misclassification. A 100-article stratified gold-set evaluation of the evidence tagger measured 46% exact-label accuracy, with binary evidence-detection precision of 96%, recall of 55%, and F1 of 0.701 (Figure 3.2). The tagger therefore behaves as a conservative detector: when it assigns an evidence label, that label is almost always

correct, but it misses roughly half of valid evidence-bearing records — particularly in the clinical-other (23% recall), theoretical (5% recall), and preclinical-in-vitro (35% recall) categories. These metrics should be used to calibrate how absence claims are interpreted throughout.

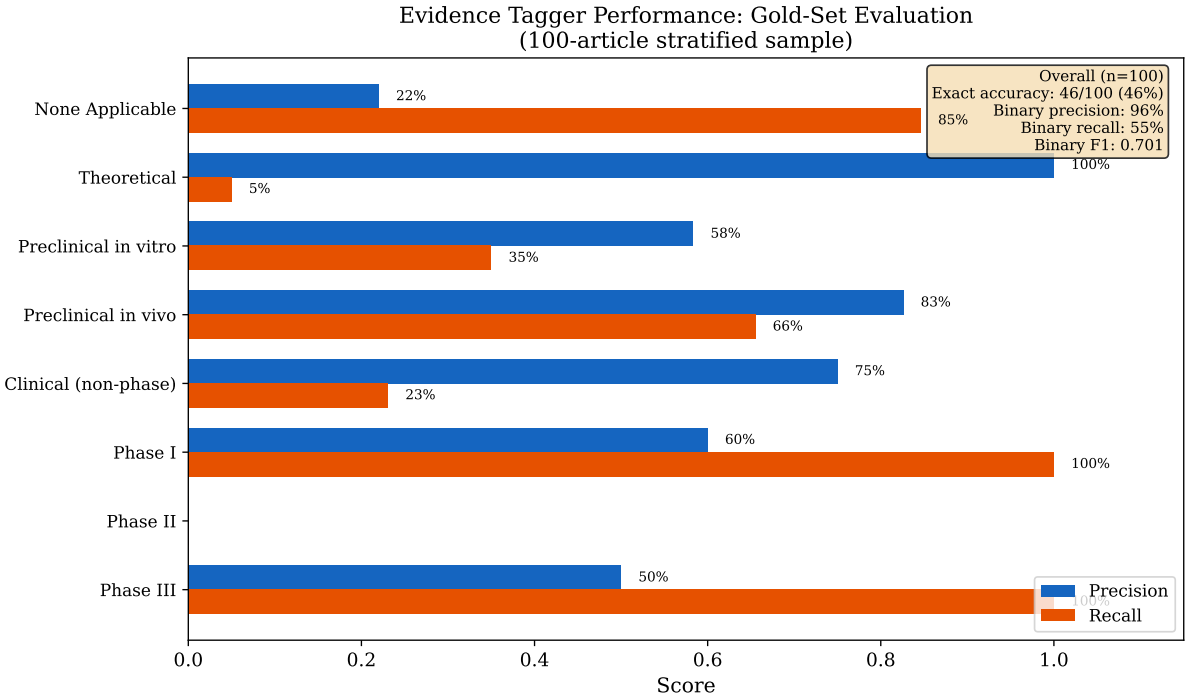

Figure 3.2: Evidence tagger performance: gold-set evaluation (100-article stratified sample).

Short or ambiguous terms may generate false positives, and articles using non-standard terminology may be missed. Second, the archive was constructed from PubMed queries only and does not capture every relevant publication, particularly articles in non-English languages, preprints not yet indexed in PubMed, or studies using unconventional terminology for established mechanisms. Third, the quantitative corpus analyzed here is restricted to the 4,830 records with local full text; although this improves traceability and reproducibility, it also excludes 5,584 abstract-only records that may materially change article-count-based field comparisons. A second-order consequence is that some landmark papers are missing from the local full-text archive even when they are clearly field-defining at the PubMed level; radioligand therapy is the clearest current example because the VISION trial (PMID 34161051) is not locally present. Fourth, evidence-level classification was based on keyword detection plus publication-type metadata (e.g., "Clinical Trial, Phase III", "phase 3", "pilot study", "case report", "in vivo") rather than independent trial registry verification; importantly, evidence-tier coverage remains incomplete even after adding a conservative ‘clinical-other‘ bucket for non-phase patient studies. Claims about the absence of clinical trials for specific mechanisms should therefore be inter-

preted as "not detected in our local keyword-based analysis" rather than definitive statements. Fifth, multi-mechanism co-occurrence counts reflect articles tagged with multiple mechanism keywords, which includes both intentional combination studies and reviews that discuss multiple mechanisms without testing them in combination; a corpus-wide designed-combination classifier (Section 3.13) measures the designed-combination rate at 19.7%, substantially below the 40% co-occurrence rate. Sixth, the 19-mechanism taxonomy has not been tested for sensitivity: collapsing the 7 physical/energy categories into 2-3 broader groups (physical ablation, bioelectric modulation, field-based therapy) would reduce the number of "zero publication gaps" and could change the gap analysis conclusions. Our findings should be interpreted as taxonomy-dependent until sensitivity analyses with alternative groupings are performed. Seventh, our current simulation framework does not fully model in vivo lipid remodeling, including SCD-driven MUFA enrichment and other backup ferroptosis defense programs, so computational support for ferroptosis-based interventions remains stronger for in vitro-like assumptions than for established human tumor biology. Finally, article counts serve as a proxy for research intensity but do not directly measure quality, reproducibility, or clinical significance, and the corpus inherits any publication biases present in the indexed literature.

3.7 Corpus Overview

The final full-text corpus comprised 4,830 articles from 803 journals, with impact metrics available for 3,739 articles via iCite. Publication volume increased 31-fold over the study period, from 38 full-text articles in 2015 to 1,167 in 2025, with 186 articles already indexed in the first quarter of 2026. Growth was non-linear, with a marked inflection point in 2020 (257 full-text articles, a 2.4-fold increase over 2019's 106 articles), coinciding with the maturation of checkpoint immunotherapy, cell therapy, and RNA-platform attention. One important revision-era lesson is that not every post-2020 rise should be read as cancer-therapy-specific signal: for example, mRNA-vaccine counts were materially inflated until supportive and infectious-vaccine spillover was removed from that tag.

The fastest-growing mechanisms by absolute article count between 2020 and 2025 were immunotherapy (+504% annual articles), nanoparticle delivery (+800%), CAR-T therapy (+423%), and sonodynamic therapy (+355%). These growth rates reflect both the maturation of established approaches and the emergence of new platform technologies.

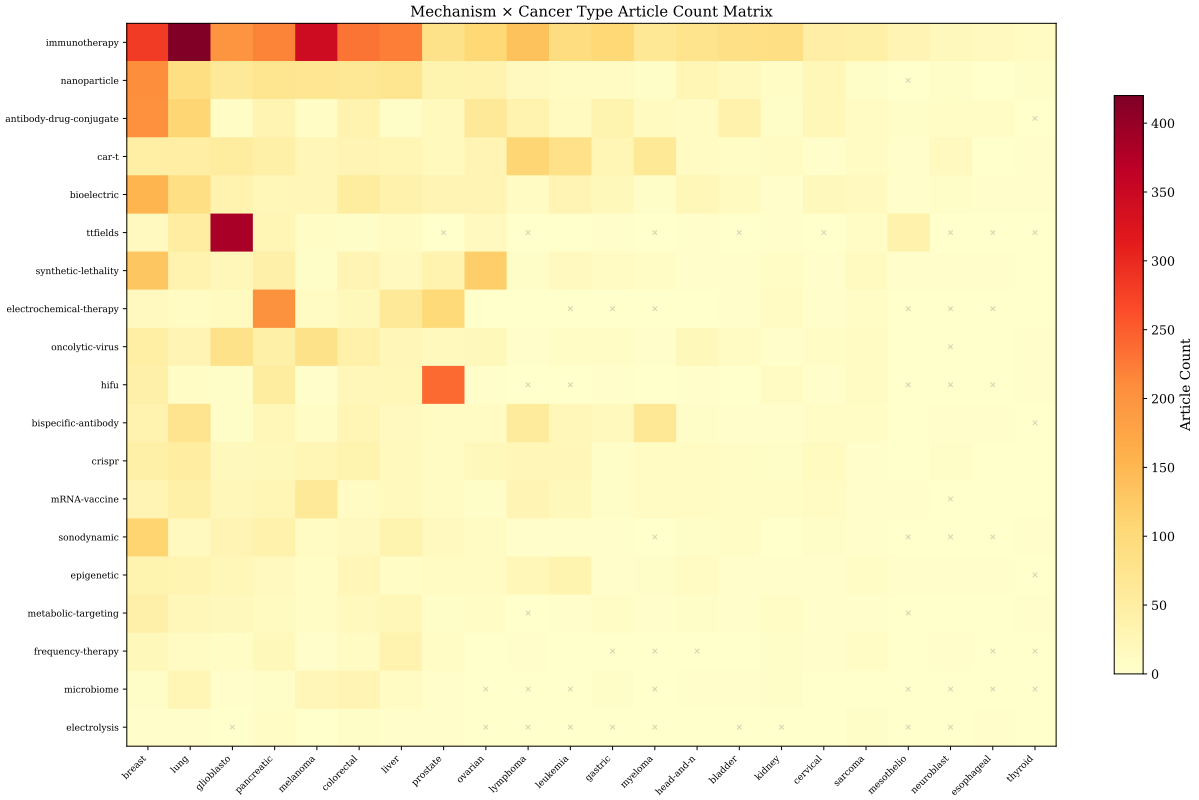

Figure 3.3: Mechanism × cancer type matrix.

3.8 Physical and Energy-Based Mechanisms

3.8.1 Tumor-Treating Fields (TTFields)

TTFields, which deliver low-intensity (1-3 V/cm), intermediate-frequency (100-300 kHz) alternating electric fields to disrupt mitotic spindle formation and organelle assembly in dividing cells O et al. [2021], represent the most clinically advanced physical therapy mechanism in our corpus. We identified 262 TTFields-tagged full-text articles across 14 of 22 cancer types. The evidence-level analysis identified 19 articles classified at the Phase III tier; however, this count still includes secondary analyses and trial-adjacent reports in addition to primary Phase III trial publications. The true number of distinct Phase III TTFields trial reports is therefore smaller than the keyword-derived total.

The evidence base is anchored by the EF-14 trial, a pivotal Phase III randomized clinical trial demonstrating that TTFields plus maintenance temozolomide significantly improved median overall survival (20.9 vs. 16.0 months, $p < 0.001$) and 5-year survival (13% vs. 5%) in newly diagnosed glioblastoma R et al. [2017]. This trial, with 2,494 citations, is the fifth-highest-impact article in our entire corpus (RCR 69.5, 100th percentile). An earlier randomized trial R et al. [2015] (1,203 citations) established the foundation for FDA approval of the Optune device.

Beyond glioblastoma, TTFields are being investigated in mesothelioma (38 articles),

lung cancer (51 articles), pancreatic cancer (27 articles), and ovarian cancer (15 articles). The LUNAR Phase III trial demonstrated improved overall survival in metastatic non-small-cell lung cancer (NSCLC) when TTFields were added to standard systemic therapy T et al. [2023]. A comprehensive review characterizes TTFields as a distinct fourth modality of cancer treatment alongside surgery, radiation, and pharmacotherapy O et al. [2021].

TTFields have zero or near-zero publications in prostate cancer, lymphoma, myeloma, bladder cancer, and cervical cancer in our corpus, though some of these gaps may reflect search query design rather than genuine absence.

3.8.2 High-Intensity Focused Ultrasound (HIFU)

HIFU uses convergent acoustic energy to achieve thermal ablation of tumor tissue at a defined focal point without incision. Our full-text corpus contains 81 HIFU articles across 12 cancer types. The dominant application is prostate cancer (29 full-text articles), where HIFU is established as a focal therapy option E et al. [2015]; S et al. [2018]. Notably, a multicentric study reported 5-year outcomes of focal HIFU for clinically significant nonmetastatic prostate cancer S et al. [2018] (309 citations), demonstrating favorable oncologic control with preserved quality of life.

HIFU research in breast cancer (42 articles), pancreatic cancer (50 articles), and liver cancer (22 articles) is expanding. A systematic review positions HIFU alongside radiofrequency ablation, microwave ablation, and cryoablation as viable alternatives to surgical resection for small tumors van dVEMF et al. [2021].

3.8.3 Sonodynamic Therapy

Sonodynamic therapy (SDT) exploits low-intensity ultrasound to activate sonosensitizer molecules that generate reactive oxygen species (ROS) within tumor tissue. Our full-text corpus contains 187 SDT articles across 15 cancer types, making it one of the most active physical modalities in the energy-based subset. However, SDT faces significant translational challenges. Our keyword-based evidence classification detected only 1 Phase I article in the local full-text corpus, with the balance of the literature remaining heavily preclinical. Broader PubMed verification suggests more early clinical activity may exist outside the locally indexed full-text set, but within the reproducible corpus analyzed here SDT remains overwhelmingly a preclinical field with no detected Phase II or Phase III full-text signal.

The preclinical literature is large and growing. Landmark studies have demonstrated nanoenzyme-augmented SDT with catalytic tumor oxygenation P et al. [2018] (571 citations), graphene-augmented sonocatalytic tumor eradication C et al. [2017] (300 citations), and piezotronic effect-augmented sonosensitizers Y et al. [2022a] (218 citations).

The convergence of SDT with nanoparticle delivery and immunotherapy is particularly active, with emerging evidence for SDT-based immunogenic cell death M et al. [2021a]; Y et al. [2023].

A distinctive molecular signature emerges from our full-text corpus analysis that keeps SDT relevant to the ferroptosis conversation even after broadening the manuscript frame. SDT is strongly associated with **ferroptosis** language — iron-dependent, lipid peroxidation-mediated cell death — with 36 full-text articles linking SDT to ferroptosis, compared to 0 for TTFields, 1 for HIFU, and 3 for IRE in the frozen corpus. More specifically, 79 SDT full-text articles (42% of the SDT corpus) mention the glutathione/GPX4 axis, underscoring how centrally redox buffering appears in this literature. The proposed causal chain — ultrasound-activated ROS generation → glutathione depletion → GPX4 inactivation → lipid peroxidation → ferroptosis → DAMP release → immunogenic cell death — is supported at each individual link by published preclinical data Y et al. [2021a]; Q et al. [2021]; Y et al. [2021b]; Q et al. [2018]. What this pattern does **not** establish is clinical distinctiveness over PDT; that comparison is addressed in the Discussion.

The primary barriers to clinical translation include the lack of standardized sonosensitizer formulations approved for human use, the absence of validated ultrasound dosimetry protocols for SDT (as distinct from diagnostic or HIFU applications), and the challenge of demonstrating SDT superiority or complementarity to established modalities in registrational trials. Breast cancer is the most studied SDT target in the full-text corpus (42 articles), followed by liver cancer (19 articles) and pancreatic cancer (12 articles).

3.8.4 Frequency Therapy and Electrolysis

Frequency therapy — encompassing pulsed electromagnetic fields (PEMF), radiofrequency ablation in immunomodulatory contexts, and resonance-based approaches — is represented by 71 full-text articles across 11 cancer types. This is a nascent field with mechanistic diversity that complicates systematic evaluation. A foundational review M et al. [2016a] (113 citations) described mechanisms of PEMF in oncology, including disruption of mitotic spindle formation, induction of apoptosis, and immunomodulatory effects.

Electrolysis as a direct antitumor mechanism is the smallest category in our full-text corpus (11 articles across only a handful of cancer contexts), but includes innovative work on amorphous liquid metal electrodes for conformable electrochemical tumor therapy X et al. [2017] (114 citations) and platinum nanoparticle-enabled electrodynamic therapy T et al. [2019] (211 citations).

3.8.5 Electrochemical Therapy

Electrochemical therapy (EChT), dominated by irreversible electroporation (IRE), is represented by 185 full-text articles across 14 cancer types. Pancreatic cancer (61 articles) and prostate cancer (30 articles) are the primary applications. A foundational paper introduced the concept of tissue ablation through irreversible electroporation RV et al. [2005] (1,241 citations), and subsequent clinical development has demonstrated safety and feasibility in locally advanced pancreatic cancer RCG et al. [2015] (400 citations) and localized prostate cancer MJV et al. [2016]; MJ et al. [2018].

A pivotal development is the combination of IRE with immunotherapy: high-frequency IRE has been shown to induce immunogenic cell death Ringel-Scaia et al. [2019] (168 citations), and emerging data demonstrate synergy between IRE and checkpoint blockade plus TLR7 stimulation JSS et al. [2019] (111 citations). The IRE-immunotherapy convergence (45 articles) represents a growing axis of investigation. However, lymphoma, leukemia, gastric cancer, and myeloma have zero EChT publications despite established clinical experience in other solid tumors.

3.9 Bioelectric Modulation

Bioelectric research encompasses membrane potential manipulation, ion channel modulation, voltage-gated signaling in cancer biology, and electroporation-based drug delivery. Our full-text corpus contains 182 articles across 19 cancer types, with broad conceptual range but limited direct clinical antitumor translation.

A landmark Cell paper established the conceptual framework for bioelectric signaling as a reprogrammable circuit underlying embryogenesis, regeneration, and cancer M et al. [2021b] (463 citations), arguing that endogenous bioelectric patterns constitute an informational layer that can be therapeutically manipulated. Earlier work on endogenous voltage gradients controlling growth and form M et al. [2017] (289 citations) provided the biophysical foundation. The identification that mitochondrial membrane potential marks cells with enhanced stemness M et al. [2016b] (370 citations) has implications for targeting cancer stem cell populations.

Despite this conceptual richness, translational progress toward direct antitumor bioelectric interventions has been limited. In the current evidence-tier rerun, no phase-labeled trial evidence was detected for direct bioelectric antitumor therapy. The practical implication is that the bioelectric literature remains conceptually rich but clinically immature within this corpus.

3.10 Immune-Based Approaches

3.10.1 Checkpoint Immunotherapy

Checkpoint immunotherapy is the dominant mechanism in our full-text corpus, with 2,297 articles across all 22 cancer types. This reflects its established clinical role across multiple tumor types. The highest concentrations are in lung cancer (250 articles), melanoma (207 articles), breast cancer (162 articles), pancreatic cancer (138 articles), colorectal cancer (136 articles), liver cancer (133 articles), and glioblastoma (117 articles).

Key landmark articles include a comprehensive review of PD-1/PD-L1 and CTLA-4 checkpoint inhibitors Y et al. [2022b] (1,191 citations), the comprehensive management review establishing temozolomide plus TTFields as the standard of care in glioblastoma and surveying immunotherapy as a promising but significantly challenged area due to the blood-brain barrier and unique immune microenvironment AC et al. [2020] (1,989 citations), and combination immunotherapy for hepatocellular carcinoma reviewing how approximately 30% of HCC exhibits intrinsic resistance to single-agent ICIs and how combination strategies (anti-PD-1/PD-L1 with anti-angiogenics or dual checkpoint blockade) have improved first-line outcomes L et al. [2023] (310 citations). STAT3 targeting as a convergence point for oncogenic signaling and immunomodulation has been extensively reviewed S et al. [2020a] (1,011 citations).

The most active convergence axes for immunotherapy in the full-text corpus include combinations with CAR-T (417 articles), oncolytic viruses (262 articles), nanoparticle delivery (193 articles), mRNA vaccines (131 articles), CRISPR (166 articles), bispecific antibodies (139 articles), metabolic targeting (127 articles), microbiome modulation (100 articles), ADCs (78 articles), SDT (53 articles), and TTFields (47 articles). Immunotherapy remains the central hub of the convergence network.

3.10.2 CAR-T Cell Therapy

CAR-T therapy is represented by 474 full-text articles across all 22 cancer types. The literature has evolved from early hematologic successes in B-cell leukemia and lymphoma S et al. [2020b] (1,400 citations) toward aggressive expansion into solid tumors.

Key developments include: IL-15-armored GPC3-targeted CAR-T cells for solid cancers D et al. [2025] (170 citations), representing a next-generation approach to overcoming the immunosuppressive tumor microenvironment; Claudin18.2-specific CAR-T cells showing promising Phase I results in gastrointestinal cancers C et al. [2022] (546 citations); and IL-10-expressing CAR-T cells that resist dysfunction and mediate durable clearance of solid tumors Y et al. [2024a] (228 citations), demonstrating that cytokine engineering of CAR-T cells can overcome exhaustion.

CRISPR-based engineering of CAR-T cells is a rapidly growing convergence axis (95

full-text articles), exemplified by RASA2 ablation enhancing antigen sensitivity J et al. [2022a] (207 citations) and adenosine A2A receptor deletion enhancing efficacy L et al. [2021] (199 citations). The allogeneic CRISPR-Cas9-engineered CAR-T therapy CTX130, targeting CD70 in T-cell lymphoma, has demonstrated safety in Phase I SP et al. [2025] (58 citations), representing a milestone for gene-edited cell therapies. CAR-T combined with oncolytic viruses (40 articles) and bispecific antibodies (30 articles) represents additional emerging convergence zones.

3.10.3 Bispecific Antibodies

Bispecific antibodies, engineered to simultaneously bind two distinct targets (typically a tumor antigen and an immune effector molecule), are represented by 247 full-text articles across 20 of 22 cancer types. Myeloma is a particularly active area (22 articles), with real-world data on teclistamab demonstrating clinical utility in relapsed/refractory disease C et al. [2024] (94 citations). Lung cancer applications are expanding, with KN046, a PD-L1/CTLA-4 bispecific, showing Phase II results in combination with chemotherapy for first-line metastatic NSCLC Y et al. [2024b] (38 citations), and the EGFRxHER3 bispecific izalontamab reporting first-in-human results J et al. [2025].

Bispecific antibody convergence with ADCs (42 articles) is an active area combining target engagement with cytotoxic payload delivery.

3.10.4 mRNA Cancer Vaccines

mRNA cancer vaccines represent one of the more dynamic but also more taxonomy-sensitive areas in the full-text corpus: 179 articles across 20 cancer types after tightening the mechanism tag to therapeutic cancer-vaccine studies. Earlier drafts materially overcounted this lane because supportive and infectious-vaccine papers in cancer cohorts bled into the mechanism definition. The field was nevertheless catalyzed by the COVID-19 mRNA platform moment, which demonstrated the feasibility of rapid, scalable mRNA production and lipid nanoparticle delivery GT et al. [2022] (279 citations). A comprehensive Lancet Oncology review established the clinical landscape CL et al. [2022] (470 citations), documenting the depth of ongoing therapeutic development.

The landmark article in this category is the Nature report on personalized RNA neoantigen vaccines stimulating T-cell responses in pancreatic ductal adenocarcinoma LA et al. [2023] (1,153 citations, RCR 90.3, 100th percentile) —the second-highest-impact article in our entire corpus. This Phase I trial of autogene cevumeran, an individualized neoantigen vaccine based on uridine mRNA-lipoplex nanoparticles, demonstrated that personalized vaccination could elicit measurable T-cell responses against patient-specific neoantigens in one of the most treatment-resistant cancers.

The mRNA vaccine field still intersects strongly with immunotherapy (131 co-tagged full-text articles) and nanoparticle delivery (47 co-tagged articles), reflecting the inherent dependence of mRNA vaccines on both immune activation and delivery technology. In the current evidence-tier table, the mechanism reaches Phase II, but its local full-text signal remains review-heavy and only partially covered by patient-study tagging. That means the field is clearly active, but mechanistic article counts should not be mistaken for mature comparative clinical evidence. A recent clinical advances review documents the current pipeline across melanoma, glioblastoma, lung, breast, and prostate cancers AV et al. [2025] (74 citations). Lipid nanoparticle-mediated lymph node-targeting delivery of mRNA cancer vaccines has been shown to elicit robust CD8+ T-cell responses J et al. [2022b] (371 citations).

3.10.5 Oncolytic Viruses

Oncolytic virotherapy is represented by 378 full-text articles across 19 cancer types. Glioblastoma is the most studied target (43 articles), followed by melanoma (51 articles), where T-VEC (talimogene laherparepvec) established the precedent for oncolytic virus approval. A comprehensive Nature Reviews Clinical Oncology review characterizes the field's progress and challenges SZ et al. [2023] (489 citations), noting that oncolytic viruses offer selective replication in tumor cells, delivery of multiple transgene payloads, induction of immunogenic cell death, and promotion of antitumor immunity.

A notable 2025 Cell paper reported a hyperacute rejection-engineered oncolytic virus designed for interventional clinical trials in refractory cancer patients L et al. [2025] (67 citations), addressing the critical limitations of intravenous safety and inherent immune neutralization. In hepatocellular carcinoma, the oncolytic virus VG161 showed activity in a Nature-published Phase I study Y et al. [2025] (53 citations). The emerging field of OV-based cancer immunotherapy has been comprehensively reviewed in Trends in Cancer R et al. [2023] (264 citations), positioning oncolytic viruses as multi-faceted tumor killers that combine direct lysis with immune potentiation.

The strongest convergence axis is oncolytic virus-immunotherapy (262 full-text articles), consistent with the mechanistic rationale that viral oncolysis generates immunogenic cell death that may synergize with checkpoint blockade SZ et al. [2023]; R et al. [2023]. Combinations with CAR-T (40 articles) and nanoparticle delivery remain emergent.

3.11 Precision Molecular Approaches

3.11.1 CRISPR Gene Editing

CRISPR-based approaches are represented by 331 full-text articles across 20 cancer types, with no Phase II or Phase III articles detected by our automated classification, placing CRISPR among the most preclinical-heavy mechanisms in our analysis. Despite this, the conceptual impact is enormous: CRISPR screens have defined unified hallmarks of cancer cell-autonomous immune evasion, providing a systematic map of therapeutic targets A et al. [2026].

Key convergence axes include CRISPR-immunotherapy (166 articles), CRISPR-CAR-T (95 articles), CRISPR-synthetic lethality (50 articles), and CRISPR-nanoparticle (29 articles). The engineering of next-generation CAR-T cells with CRISPR-Cas9 has been extensively reviewed A et al. [2022] (314 citations), and clinical milestones include non-viral precision T-cell receptor replacement for personalized cell therapy SP et al. [2023] (256 citations) and the COBALT-LYM Phase I trial of CRISPR-edited allogeneic CAR-T for T-cell lymphoma SP et al. [2025] (58 citations).

Precision nanomedicine integrating CRISPR with nanoparticle delivery for tumor microenvironment modulation represents a growing area H et al. [2025a] (101 citations).

3.11.2 Synthetic Lethality

Synthetic lethality — the paradigm in which loss of two genes is lethal while loss of either alone is viable —is represented by 367 full-text articles across 21 cancer types. PARP inhibitors, the clinical embodiment of this concept, have achieved broad clinical use in breast cancer (69 articles), ovarian cancer (73 articles), pancreatic cancer (25 articles), and prostate cancer (24 articles). A pan-tumor review documented the expanding role of PARP inhibitors across multiple malignancies CC et al. [2025] (21 citations).

The concept of synthetic lethality as an engine for cancer drug target discovery has been comprehensively reviewed A et al. [2020] (458 citations), and CRISPR-synthetic lethality combinations (63 articles) represent a powerful approach for systematic identification of new synthetic lethal pairs. Transcription-replication conflicts have been identified as a mechanistic basis for PARP inhibitor sensitivity M et al. [2024] (160 citations), expanding the theoretical framework beyond BRCA mutations.

Our corpus detected zero synthetic lethality-myeloma co-tagged articles; however, independent PubMed verification reveals at least 11 publications on PARP inhibitors in multiple myeloma, including early-phase clinical data on veliparib combined with bortezomib. This gap is therefore an artifact of our search query design rather than a genuine absence of research. The myeloma-PARP axis is under-researched relative to breast or ovarian cancer applications, but it is not unexplored.

3.11.3 Antibody-Drug Conjugates (ADCs)

ADCs are represented by 284 full-text articles across 21 cancer types. The field is dominated by breast cancer applications (92 articles), lung cancer (48 articles), and ovarian cancer (29 articles). The changing treatment landscape of EGFR-mutant NSCLC has been shaped by ADC development alongside bispecific antibodies F et al. [2025] (106 citations).

ADC convergence with immunotherapy (160 articles) is well established, and combinations with CAR-T (42 articles) and bispecific antibodies (42 articles) are emerging.

3.11.4 Epigenetic Therapy

Epigenetic approaches (DNA methylation inhibitors, histone deacetylase inhibitors, bromodomain inhibitors, and related agents) are represented by 183 full-text articles across 20 cancer types. A landmark Cell Death Discovery review established the current state of clinical translation X et al. [2024] (223 citations), and a Signal Transduction and Targeted Therapy paper provided a comprehensive framework for epigenetic regulation in the tumor microenvironment J et al. [2023] (322 citations).

The most impactful single finding in this category is the Nature paper demonstrating that NBS1 lactylation is required for efficient DNA repair and chemotherapy resistance H et al. [2024] (413 citations, RCR 56.2), linking metabolic reprogramming (the Warburg effect) to epigenetic modification of the DNA damage response. This exemplifies how mechanistic convergence at the molecular level can reveal new therapeutic vulnerabilities.

Epigenetic-immunotherapy convergence (78 articles) is the dominant combination.

3.12 Delivery and Microenvironment

3.12.1 Nanoparticle Delivery

Nanoparticle-based delivery systems constitute the second-largest mechanism category in our full-text corpus: 515 articles across 20 cancer types. An important distinction must be drawn between established and experimental nanoplatforms. Several nanomedicines have achieved FDA approval, including liposomal doxorubicin (Doxil, 1995), nab-paclitaxel (Abraxane, 2005), liposomal irinotecan (Onivyde, 2015), and liposomal daunorubicin-cytarabine (Vyxeos, 2017) MD et al. [2023]. These approved nanomedicines are typically described by drug name rather than "nanoparticle" in the literature and are therefore underrepresented in our corpus. The translational gap identified here primarily applies to novel experimental nanoplatforms — metal, polymer, and inorganic nanoparticles — which remain overwhelmingly preclinical in our full-text corpus. This distinction between proven liposomal/albumin-bound formulations and the broader experimental

nanoparticle literature is critical: the former demonstrate that nanomedicine can succeed clinically, while the latter face substantial barriers to replicating that success.

The nanoparticle literature spans gold, silver, and iron oxide nanoparticles P et al. [2025] (88 citations), liposomal formulations MD et al. [2023] (215 citations), lipid nanoparticles for mRNA delivery J et al. [2022b] (371 citations), and precision nanomedicine for tumor microenvironment modulation H et al. [2025a] (101 citations). The convergence with immunotherapy (193 articles) and sonodynamic therapy (116 articles) is extensive, as nanoparticles serve as both therapeutic agents and delivery vehicles for other modalities.

The fundamental translational barriers include: (a) heterogeneous nanoparticle pharmacokinetics and biodistribution that are difficult to predict from preclinical models; (b) manufacturing scalability and reproducibility challenges; (c) regulatory uncertainty around the classification and approval pathway for nanoparticle-drug combinations; and (d) the enhanced permeability and retention (EPR) effect, which is robust in rodent tumors but inconsistent in human cancers. Breast cancer (172 articles), lung cancer (76 articles), and liver cancer (66 articles) are the primary targets. Mesothelioma has zero nanoparticle publications despite the appeal of local delivery to pleural surfaces.

3.12.2 Metabolic Targeting

Metabolic targeting — exploitation of altered cancer cell metabolism, particularly the Warburg effect, glutamine addiction, and lipid metabolism dependencies —is represented by 274 full-text articles across 20 cancer types. A comprehensive Nature Reviews Drug Discovery article ZE et al. [2022] (1,127 citations) chronicles the hundred-year journey from Warburg's discovery to contemporary precision metabolic oncology.

The convergence of metabolic targeting with immunotherapy (127 articles) reflects the recognition that metabolic reprogramming of the tumor microenvironment suppresses immune function and that metabolic interventions can restore immunocompetence. The landmark demonstration that glycolysis modulation can enhance checkpoint immunotherapy C et al. [2023] (341 citations) illustrates this principle. Glutamine addiction as a therapeutic target was comprehensively reviewed in a highly cited Trends in Biochemical Sciences paper DR et al. [2010] (1,747 citations).

Metabolic targeting has limited representation in lymphoma within our corpus, though broader PubMed searches suggest this gap partly reflects our search query design.

3.12.3 Microbiome Modulation

Microbiome modulation is the most recently emergent mechanism in our analysis: 109 full-text articles across 11 cancer types. The field has been catalyzed by the recognition that gut microbiota composition influences checkpoint immunotherapy response, as

comprehensively reviewed in Nature Reviews Cancer MR et al. [2022] (258 citations) and Nature Medicine EM et al. [2022] (467 citations).

Fecal microbiota transplantation (FMT) combined with checkpoint inhibitors has emerged as a clinical strategy, with a Gut Microbes review characterizing optimization approaches A et al. [2025a] (56 citations) and a meta-analysis evaluating FMT-ICI combinations A et al. [2025b] (23 citations). The DIET randomized trial investigated dietary intervention to modulate microbiome composition in melanoma patients receiving immunotherapy RM et al. [2024] (21 citations).

Microbiome-immunotherapy convergence (100 articles) dominates the microbiome literature in our corpus.

3.13 Cross-Mechanism Convergence: Where Synergies Emerge

Of the 4,830 articles in our full-text corpus, 1,952 (40%) are tagged with two or more therapeutic mechanism keywords, and 396 carry three or more tags. To distinguish deliberate combination studies from incidental co-mentions, we applied a corpus-wide designed-combination classifier using explicit combination/synergy language and study-type signals. Of the 1,952 multi-mechanism articles, 385 (19.7%) are classified as designed combination studies — 60 with clinical patient-study signal and 325 preclinical — while 622 (31.9%) are co-mentions only and 945 (48.4%) are reviews or perspectives spanning multiple lanes (Figure 3.4). An earlier 50-article manual subsample had estimated ∼10% experimental combinations; the automated classifier yields a higher rate primarily because it captures preclinical combination designs that the small manual sample underrepresented. The remaining co-tagged articles map the intellectual territory where researchers are connecting mechanisms conceptually — a precursor to deliberate combination design, but not equivalent to experimental evidence of synergy. We distinguish between the co-occurrence rate (40%) and the measured designed-combination rate (19.7%), and caution against conflating the two.

The convergence landscape is organized around a central hub: checkpoint immunotherapy participates in 8 of the top 10 mechanism pairs and co-occurs with every other mechanism in the corpus. The top 10 convergence pairs are:

1. **CAR-T + Immunotherapy:** 417 articles. This dominant axis reflects the recognition that CAR-T cells face significant barriers in solid tumors — including exhaustion, antigen escape, and immunosuppressive microenvironments — and that checkpoint blockade can partially rescue exhausted CAR-T cell function S et al. [2020b]. 2. **Immunotherapy + Oncolytic Virus:** 262 articles. Oncolytic viruses induce immunogenic cell death that synergizes with checkpoint blockade by converting immunologically "cold"

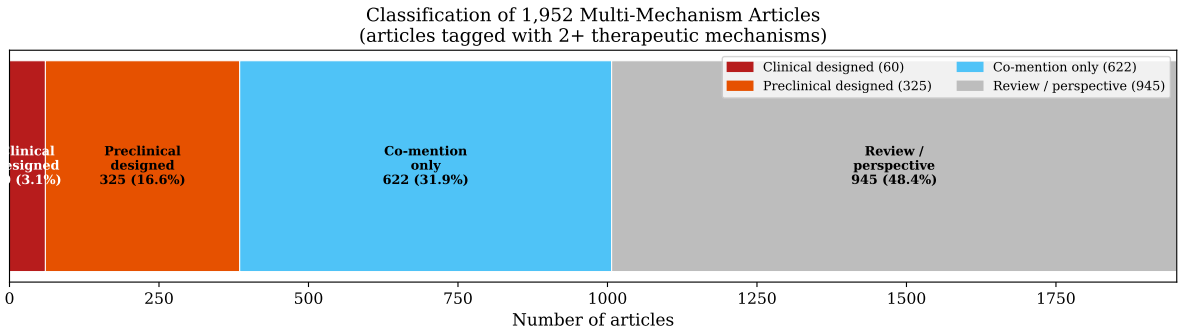

Figure 3.4: Classification of multi-mechanism articles into designed combinations, co-mentions, and reviews.

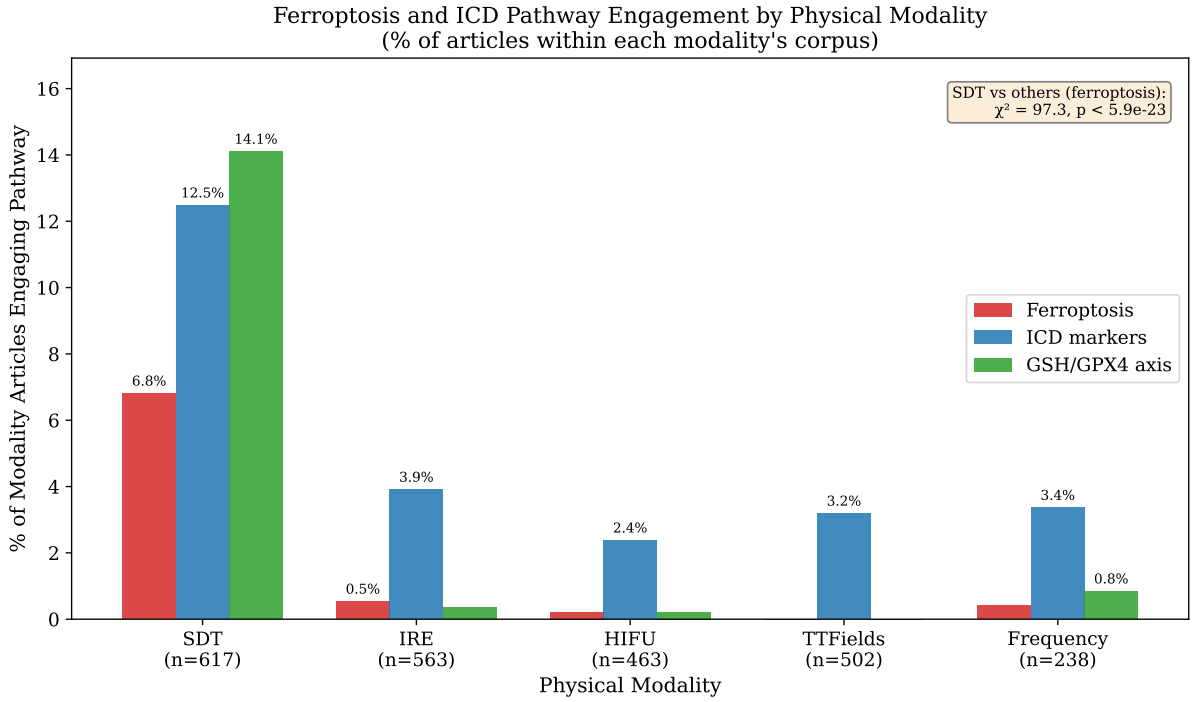

Figure 3.5: Ferroptosis engagement ($\chi^2 = 97.3$, $p < 5.9 \times 10^{-23}$).

tumors to "hot." 3. **Immunotherapy + Nanoparticle:** 193 articles. Nanoparticles serve as delivery vehicles for checkpoint inhibitor payloads, immune adjuvants, and tumor antigens for in situ vaccination. 4. **Immunotherapy + mRNA Vaccine:** 131 articles. mRNA vaccines generate neoantigen-specific T-cell responses that are amplified by checkpoint blockade. 5. **CRISPR + Immunotherapy:** 166 articles. CRISPR screens identify immune evasion mechanisms; CRISPR editing engineers superior immune cells. 6. **Bispecific Antibody + Immunotherapy:** 139 articles. Bispecific antibodies redirect immune effectors to tumors while checkpoint inhibitors release immune brakes. 7. **Immunotherapy + Metabolic Targeting:** 127 articles. Metabolic interventions can reshape the tumor microenvironment to restore immune function and improve checkpoint responses. 8. **Nanoparticle + Sonodynamic:** 116 articles. Nanoparticles function as

sonosensitizer carriers, enhancing SDT specificity and efficacy through targeted delivery and tumor microenvironment responsiveness. 9. **Immunotherapy + Microbiome:** 100 articles. Gut microbiota composition modulates systemic immune competence and checkpoint inhibitor efficacy. 10. **Bioelectric + Nanoparticle:** 34 articles. Nanoparticles enable bioelectric modulation through electroporation-mediated delivery and conductive nanoparticle-enhanced electric field effects.

Notable triple-mechanism articles include the landmark thermal ablation review integrating electrochemical therapy, frequency therapy, and immunotherapy KF et al. [2014] (1,938 citations), the personalized mRNA neoantigen vaccine study combining immunotherapy, mRNA vaccine, and nanoparticle platforms LA et al. [2023] (1,153 citations), and the CRISPR-engineered CAR-T review A et al. [2022] (314 citations).

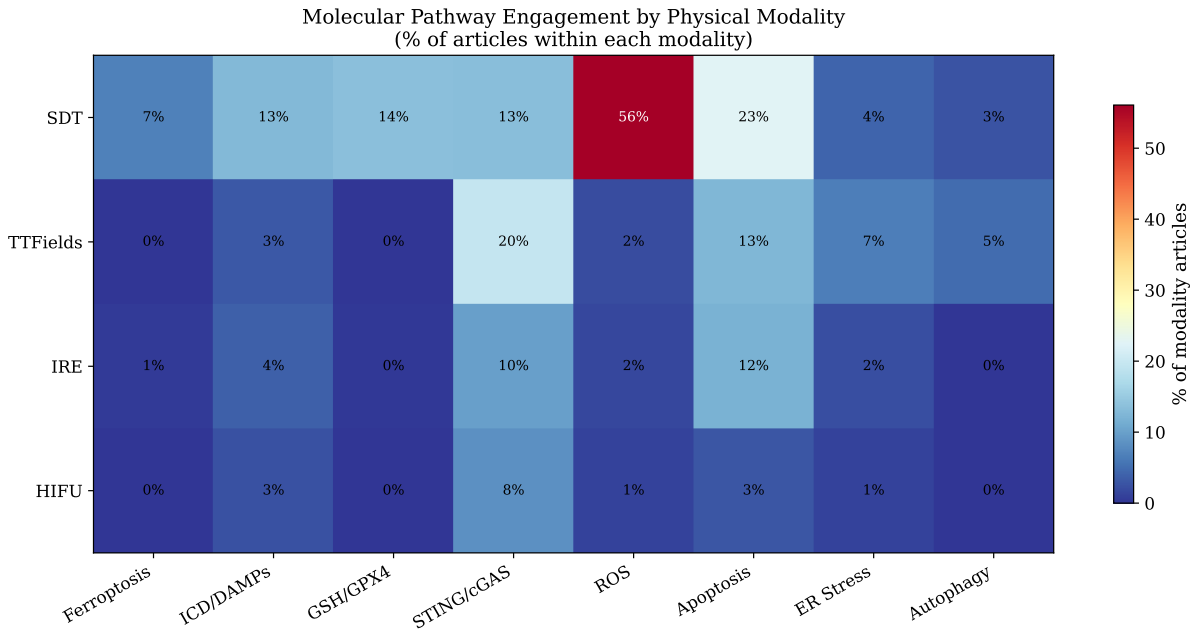

Figure 3.6: Molecular pathway engagement (normalized %).

3.14 The Mechanism-Cancer Matrix: What Works Where

The 19 x 22 mechanism-cancer matrix reveals both deep concentrations and systematic gaps.

Dominant mechanism-cancer pairs (top 5 by article count): immunotherapy-lung (250), immunotherapy-melanoma (207), TTFields-glioblastoma (204), immunotherapy-breast (162), and immunotherapy-pancreatic (138). These represent the densest full-text research axes in the current indexed corpus, though maturity of evidence varies substantially across them.

Broad-spectrum mechanisms —those studied across the widest range of cancer

types —include CAR-T therapy and checkpoint immunotherapy (22/22 cancer types), synthetic lethality and ADCs (21/22), and CRISPR, bispecific antibodies, epigenetic therapy, mRNA vaccines, metabolic targeting, and nanoparticles (20/22).

Narrow-spectrum mechanisms include electrolysis (3/22), microbiome modulation and frequency therapy (11/22 each), HIFU (12/22), TTFields and electrochemical therapy (14/22 each), and sonodynamic therapy (15/22). The narrowness of TTFields coverage is particularly noteworthy given its strong clinical evidence base.

The most intensively researched cancer types across all mechanisms are breast cancer (697 total cross-mechanism article-instances), lung cancer (593), glioblastoma (529), pancreatic cancer (428), melanoma (425), and colorectal cancer (376). The least-researched cancer types are thyroid (19), esophageal (20), mesothelioma (49), and neuroblastoma (52).

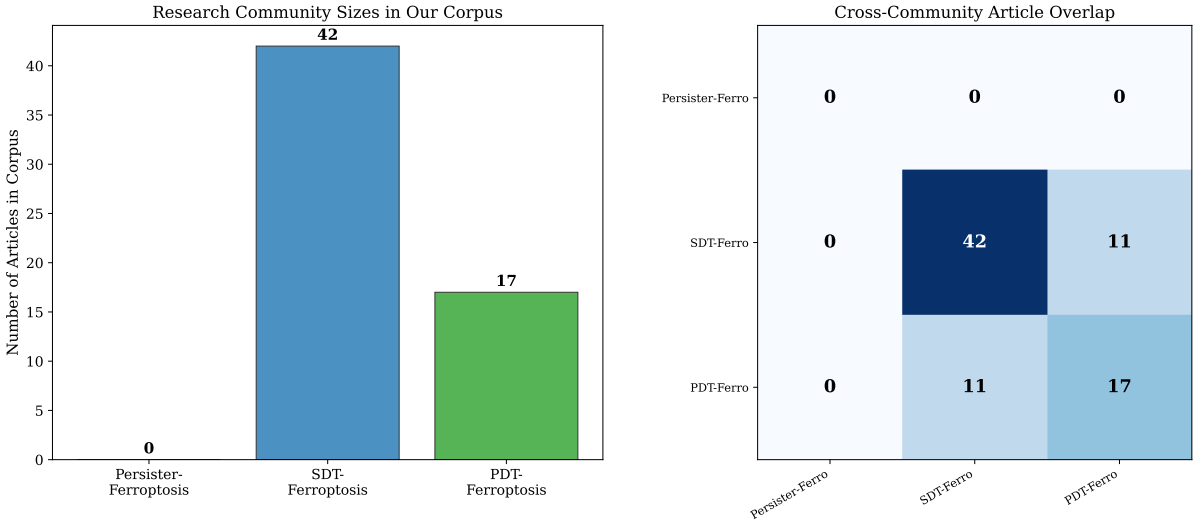

Figure 3.7: Literature disconnect between communities.

Chapter 4

Evidence Quality and Gaps

4.1 Evidence Maturity and Weighted Scoring

Evidence tags are currently populated for 2,038 of 4,830 full-text records (42.2%). Reviews and meta-analyses (1,921) and protocols (29) are intentionally left unclassified; among the remaining primary-study-like records, coverage is 2,038 of 2,880 (70.8%). The gold-set evaluation described in Section 3.6 confirms that this coverage reflects a conservative detector: tagged evidence is almost always real, but a substantial share of valid evidence goes undetected, particularly in the clinical-other, theoretical, and preclinical-in-vitro categories.

To move beyond raw counts, we computed a weighted evidence score for each mechanism that combines evidence tier (Phase III = 12 points, down to theoretical = 0.5), citation percentile (1.0-1.5x modifier), and publication recency (0.9-1.1x modifier). The resulting ranking shows immunotherapy (2,617), bispecific antibodies (687), TTFields (568), and ADCs (558) as the most evidence-mature mechanisms. Weighted scoring highlights where article volume is not matched by evidence depth: nanoparticles rank sixth by weighted evidence despite ranking second by article count, reflecting their overwhelmingly preclinical composition. Bioelectric modulation has the lowest average weight per tagged article (1.69), despite 144 tagged evidence rows, indicating high volume but minimal clinical signal.

4.2 Main Contribution: A Broader Frame Before A Narrower Hypothesis

The most important contribution of this article is not a claim that one modality should replace another. It is a change in framing. The frozen 19-mechanism corpus shows that oncology is not short of therapeutic ideas; it is short of disciplined ways to compare them. Mechanism counts, convergence maps, and evidence tiers reveal a field dominated

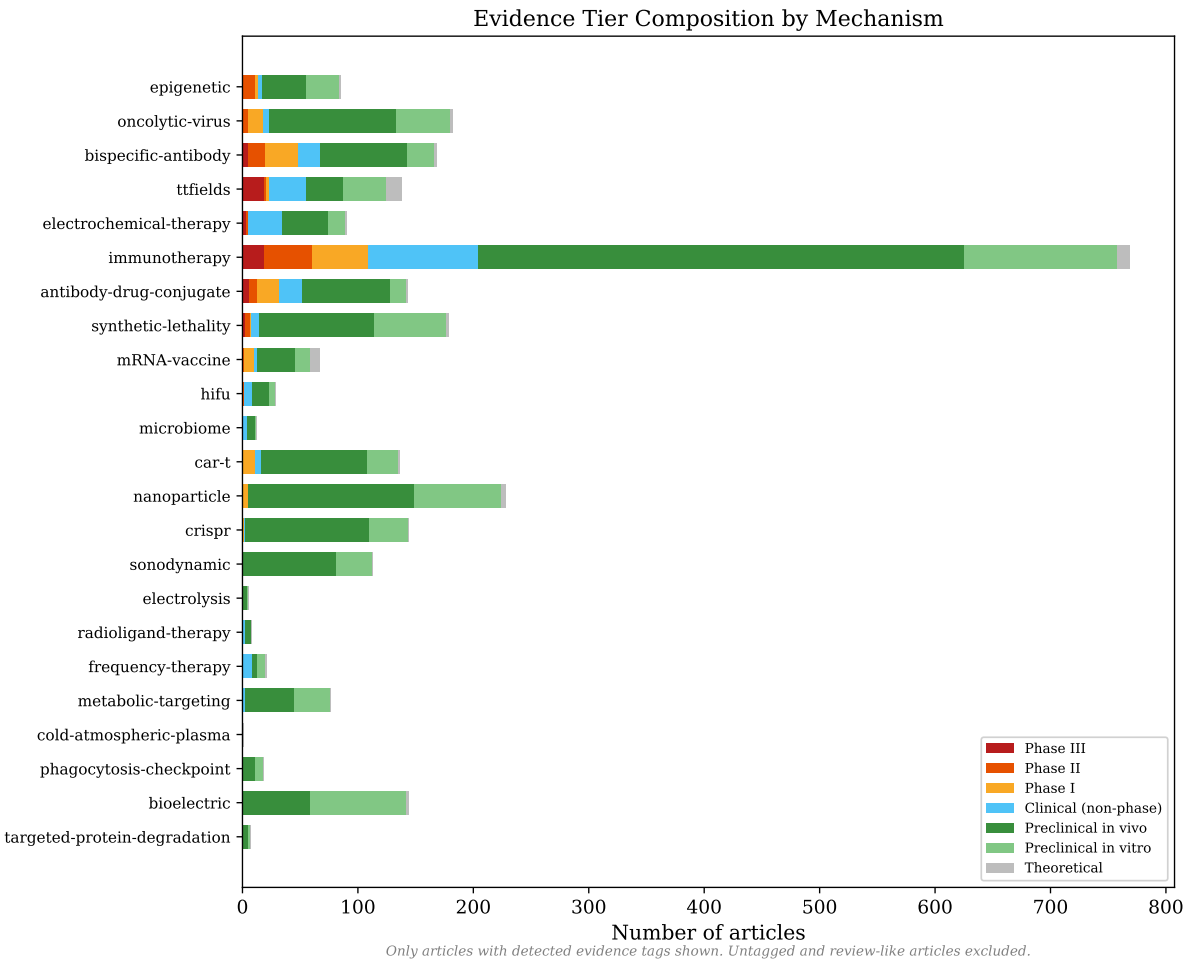

Figure 4.1: Evidence tier composition by mechanism.

by a few high-volume hubs, but they also show how quickly apparently clean conclusions can be destabilized by taxonomy choices, missing full text, incomplete evidence tagging, and tag contamination from adjacent literatures. That matters because many attractive "gaps" are not true absences, and many active literatures are not equally mature.

Within that broader frame, one narrower implication remains worth pursuing. If residual disease states are genuinely OXPHOS-skewed, lipid-peroxidation-prone, and locally targetable, then physical ROS-generating modalities may be worth evaluating alongside pharmacologic ferroptosis inducers rather than outside the conversation. That is a conditional hypothesis downstream of the landscape analysis, not the article's starting assumption.

A tissue-of-origin layer derived from existing cancer-type tags provides a coarser but more interpretable view of where research is concentrated. Of the 4,830 full-text records, 2,995 unique articles (62.0%) have at least one derived tissue category. Because a single article can study multiple cancer types and therefore map to more than one tissue group, the per-category counts below are overlapping and sum to 3,241 rather than 2,995: epithelial (2,212), neuroectodermal (667), hematologic (245), mesenchymal (77), and mesothelial

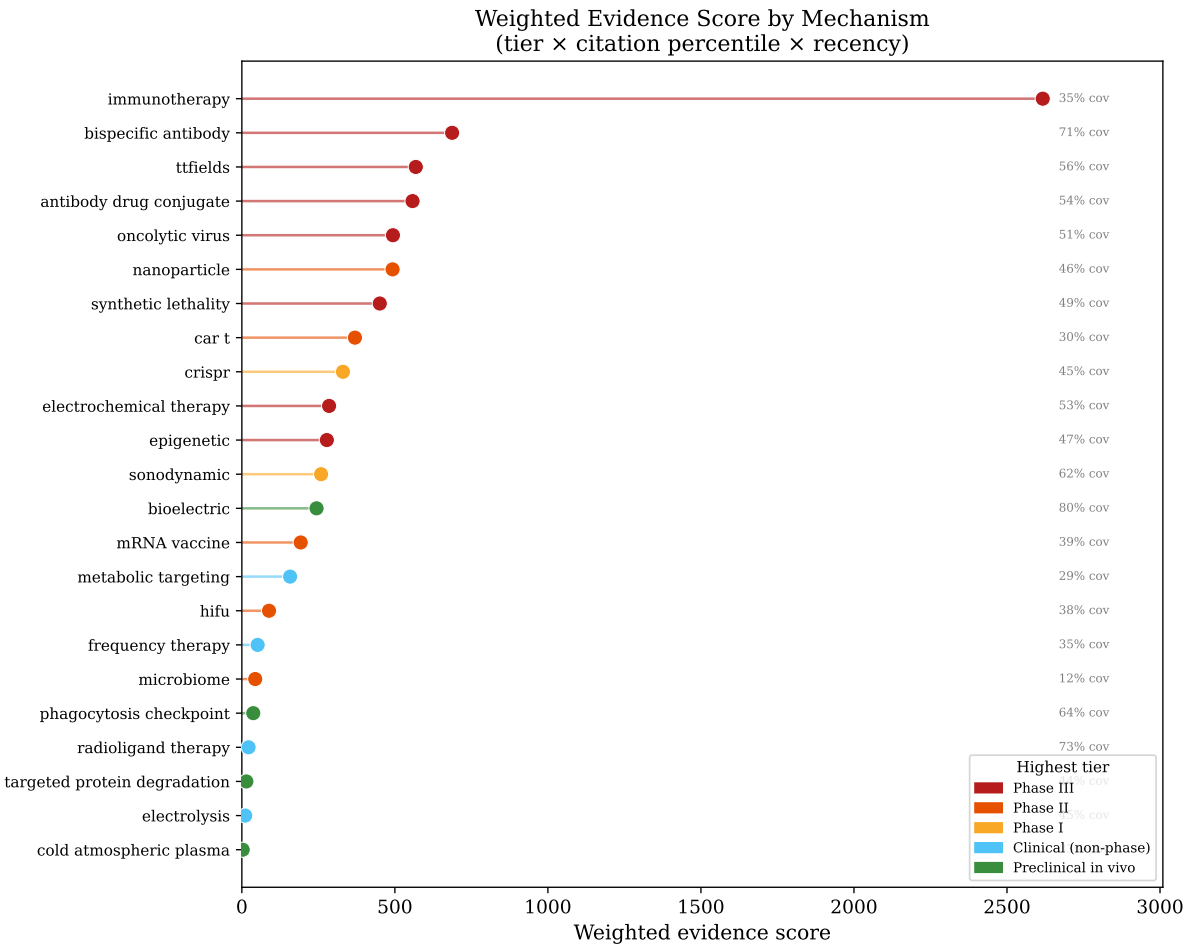

Figure 4.2: Weighted evidence score by mechanism (tier × citation percentile × recency).

(40). The physical-modality literature is overwhelmingly concentrated in epithelial and neuroectodermal contexts, with hematologic and mesothelial categories containing almost no physical-modality signal (Figure 4.3). This serves as a check against over-generalizing localized solid-tumor strategies to all cancers. The tissue layer is derived from the same cancer-type tags used in the mechanism-cancer matrix — it improves interpretation but does not fix upstream tagging errors.

4.3 What The Corpus Actually Supports About PDT, SDT, and CAP

The corpus and external cross-checking support a more modest position than our earlier framing implied.

First, PDT is the established precedent. PDT has the larger ferroptosis and ferroptosis-plus-ICD literature, more mature clinical history, and approved photosensitizers. Any argument for SDT must therefore explain why it adds something beyond a

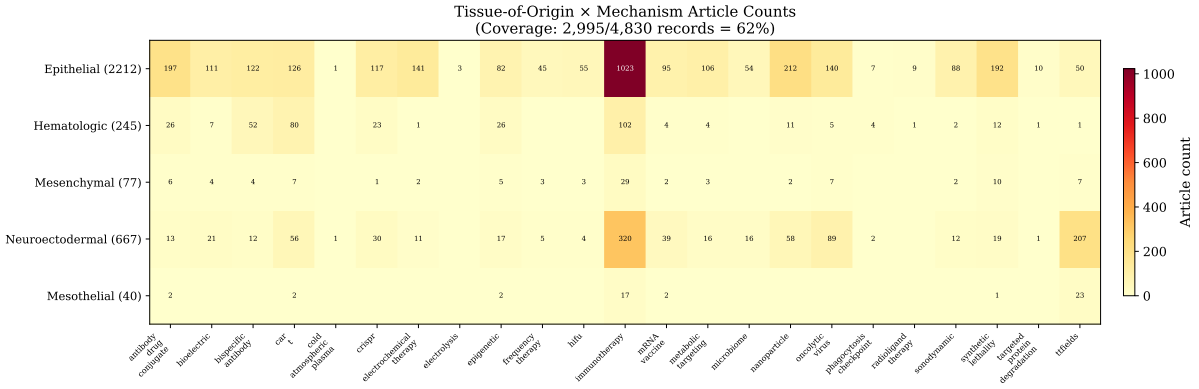

Figure 4.3: Tissue-of-origin × mechanism article counts (coverage: 62%).

depth extension of an older modality rather than pretending SDT is uniquely linked to ferroptosis or ICD.

Second, SDT's real advantage is anatomical, not conceptual. Ultrasound reaches deep soft tissue that light cannot. For pancreatic, hepatic, or pelvic disease, SDT has a genuine physical-access advantage over PDT. But the mechanism remains broadly shared: externally triggered ROS, redox-buffer exhaustion, lipid peroxidation, and potential DAMP release. The clinically relevant question is therefore where depth changes the therapeutic opportunity enough to matter.

Third, CAP is a speculative adjacent lane, not a co-equal conclusion. CAP broadens the class of externally imposed ROS therapies, but in the current local corpus it is represented by only 3 full-text articles and should therefore be treated as a hypothesis-generating comparator for accessible or locoregional settings, not as a mature pillar of the argument.

Fourth, the strongest version of the PDT/SDT claim is now conditional. Physical ROS modalities are most plausible when residual disease is locally targetable, when spatial selectivity matters more than systemic reach, and when ferroptosis-related vulnerability is expected to persist long enough for intervention. The unresolved in vivo problem is exactly where this claim can fail: SCD/MUFA remodeling, DHODH-linked backup defenses, NRF2 compensation, stromal buffering, and other context-dependent escape programs may narrow or erase the apparent window.

Part III

What The Simulations Show

Chapter 5

The Ferroptosis Engine

5.1 What the Model Represents

Before presenting what the simulations found, it is worth explaining what they model and why. The chapters that follow (Chapters 6 and 7) present specific predictions about drug combinations, tissue penetration, and tumor microenvironment resistance. Every one of those predictions depends on the model described here. Understanding the model — what it captures, what it simplifies, and what it ignores — is necessary to judge whether any individual prediction deserves experimental follow-up.

The simulation models ferroptosis, a specific type of cell death in which uncontrolled oxidation of the lipid membranes that enclose cells leads to membrane rupture and death. The process involves five key components, each represented in the model:

Glutathione (GSH) is the cell's primary antioxidant buffer — a supply of chemical that both scavenges reactive oxygen species (ROS) directly and serves as the essential cofactor that GPX4 requires to do its repair work. Without GSH, GPX4 cannot function. In the model, GSH is consumed by ROS at a rate governed by Michaelis-Menten kinetics — the more GSH available, the faster ROS is scavenged, but with diminishing returns at high concentrations.

GPX4 is the enzyme that converts dangerous lipid peroxides back to harmless products. It is the cell's primary firefighter — the last line of defense before lipid damage becomes self-sustaining. In the model, GPX4 activity is dynamic: it is degraded by oxidative stress and upregulated by the NRF2 transcription factor. RSL3, a pharmacologic ferroptosis inducer, works by covalently inactivating GPX4.

FSP1 is a backup repair pathway, independent of GPX4 and GSH. It acts like a sprinkler system — a second line of defense that operates through a different mechanism (CoQ10-mediated radical trapping rather than GSH-dependent peroxide reduction). In the persister cell phenotype, FSP1 is downregulated, leaving GPX4 as the sole repair mechanism and making these cells more vulnerable.

Lipid peroxidation (LP) represents the accumulating damage to cell membrane lipids. This is the fire. Critically, lipid peroxidation is autocatalytic: each damaged lipid molecule generates a radical that can damage adjacent lipid molecules, creating a chain reaction. In the model, once LP exceeds the cell's repair capacity, the cascade accelerates — one damaged molecule becomes two, then four, then eight. This creates a bistable switch: cells either recover (repair outpaces damage) or collapse irreversibly (damage outpaces repair).

The death threshold (10.0) is the LP level at which the cell is considered dead (a model-defined threshold, not a measured biological constant). Below this threshold, the cell can in principle recover if repair mechanisms are restored. Above it, the membrane is compromised beyond recovery. This is the point where the fire becomes uncontrollable.

Each treatment acts on this system differently. **RSL3** covalently inactivates GPX4 — it disables the firefighter, leaving only the backup sprinkler system (FSP1). **SDT and PDT** deliver large doses of exogenous ROS — they start a much bigger fire than the cell's normal metabolism produces. **FSP1 inhibitors** disable the backup sprinkler system, which is devastating when combined with GPX4 inhibition because it leaves the cell with no repair pathway at all.

5.2 Simulation Design

To test whether the proposed persister ferroptosis vulnerability is biochemically plausible, we developed a Monte Carlo simulation of the ferroptosis cascade. The model simulates the ROS → GSH depletion → GPX4/FSP1 inactivation → lipid peroxidation → cell death chain for individual cells with stochastic parameter variation, using literature-sourced biochemical values (Methods in Supplementary). We simulated 1,000,000 cells for each of 16 conditions (4 phenotypes × 4 treatments = 16,000,000 total cells).

Model features: (i) autocatalytic lipid peroxidation propagation gated by GSH/GPX4 — creating a bistable switch where cells either recover or collapse irreversibly (Porter et al., Chem Rev 2005); (ii) dynamic GPX4 regulation — degraded by oxidative stress, upregulated by NRF2 (Dodson et al., Free Radic Biol Med 2019); (iii) FSP1 as a GPX4-independent repair pathway (Bersuker et al., Nature 2019), downregulated in the persister phenotype (Higuchi et al., PMID:41481741); (iv) uncapped GSH depletion under exogenous ROS.

Calibration constraint: All phenotypes must show <2% death under Control (no treatment), reflecting the biological reality that even fragile persister cells are viable at baseline.

Results (Figure 7):

Key findings from the simulation:

1. **RSL3 selectively kills persisters (42.5%) while sparing glycolytic cells**

Table 5.1: Monte Carlo ferroptosis simulation (n=1M cells/condition).

Phenotype	**Control**	**RSL3**	**SDT**	**PDT**
Glycolytic	0.00%	0.00%	87.2%	87.2%
OXPHOS	0.04%	1.1%	99.9%	99.9%
Persister (FSP1↓)	1.2%	42.5%	100.0%	100.0%
Persister + NRF2	0.00%	0.05%	99.5%	99.5%

(0%), consistent with experimental findings (PMID:41481741). The model correctly reproduces the published biology: FSP1 downregulation leaves GPX4 as the sole lipid repair pathway, and its pharmacologic inhibition is lethal.

2. **SDT/PDT kills all cell types at high rates (87-100%)** because the exogenous ROS burst overwhelms GSH defense regardless of phenotype. This is an important finding: **physical ROS modalities are not inherently selective for persisters**. Their advantage over RSL3 is not selectivity but spatial confinement (focused ultrasound/light limits damage to the tumor volume). Any ICD advantage remains a biological hypothesis, not a result established by this model.

3. **The bistable switch is the critical mechanism.** Under SDT, GSH is catastrophically depleted (5.0 → 0.03 mM), which unlocks autocatalytic lipid peroxidation propagation. Cells that maintain GSH above the threshold recover; those that don't collapse irreversibly. This is qualitatively different from a linear dose-response.

4. **NRF2 compensation fully rescues from RSL3** (0.05% death, matching NRF2's known role in GPX4 upregulation) **but NOT from SDT** (99.5% death). The ROS burst from SDT overwhelms even NRF2-elevated GSH. This suggests that the NRF2 compensation failure mode we identified as the paper's primary risk may be less problematic for SDT than for pharmacologic ferroptosis inducers — a novel prediction from the simulation.

5. **Sensitivity analysis**: The qualitative result tested in the sensitivity suite — Persister > Glycolytic under SDT — holds in 22/22 parameter perturbation conditions (±50% on each of 11 rate constants), indicating robustness to parameter uncertainty for that specific comparison.

5.3 Where the Numbers Come From

The simulation uses approximately 30 parameters. Some are drawn from published experimental measurements: GSH concentrations in cancer cells, GPX4 catalytic rates, the Michaelis-Menten constants for antioxidant scavenging, and the physical attenuation coefficients for light and ultrasound in tissue. Others are estimated from indirect evidence: FSP1 repair rates, NRF2 upregulation kinetics, the rate at which exogenous ROS

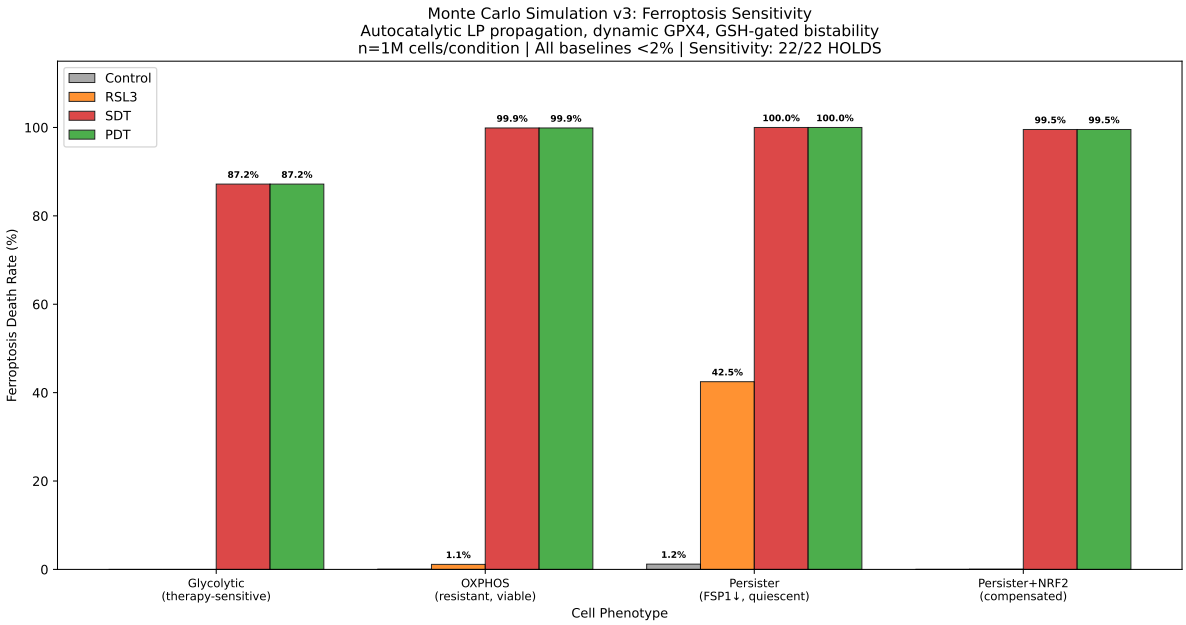

Figure 5.1: Monte Carlo simulation (1M cells/condition).

generates lipid radicals, and the stochastic variation range for each parameter.

Each cell in the simulation receives its own parameter set, drawn from a ±20% uniform random variation around the population mean. This reflects the biological reality that no two cells are identical — even genetically identical cells in culture show measurable variation in protein expression and metabolic state. By running 1,000,000 cells per condition, the simulation produces statistically robust population-level outcomes (95% confidence intervals reported for all death rates).

Five calibration constraints anchor the model to reality: (1) all phenotypes must show less than 2% death under Control conditions, reflecting baseline cell viability; (2) RSL3 must selectively kill persister cells at a rate consistent with published experimental observations; (3) the vulnerability window must close on a timescale consistent with known defense-pathway recovery kinetics. Two additional constraints verify the physics: PDT attenuation and SDT penetration must match published tissue optics and acoustics data.

5.4 What the Model Cannot Do

This is a qualitative model testing directional predictions, not a quantitative predictor of clinical outcomes. The absolute percentages (87%, 42%, etc.) depend on model-specific rate constants and should not be interpreted as clinical predictions. The model's value is in demonstrating that the proposed biochemical chain produces the expected differential sensitivity under biologically plausible parameters, and in identifying which conditions

favor which outcomes.

Specific limitations the reader should keep in mind throughout Chapters 6 and 7:

The model does not include upstream signaling — iron import, System Xc- cystine transport, and ACSL4-mediated lipid remodeling are not represented. It does not model adaptive evolution — there is no mutation, no selection over generations, no population dynamics beyond the single 180-step treatment window. Spatial drug distribution is handled separately in Chapter 6's spatial simulation, not in the baseline model. Immune response is handled separately in Chapter 7's immune coupling layer.

All fields (oxygen, pH, drug concentration) are steady-state — they are computed once based on distance from the tumor edge, not dynamically updated as cells consume resources or die. In a real tumor, blood flow fluctuates, metabolites are consumed and replenished, and the microenvironment evolves over hours to days. The steady-state approximation captures the spatial gradient but not the temporal dynamics.

The model is a hypothesis-testing tool, not a digital twin of a tumor. It identifies which biological conditions matter for ferroptosis sensitivity, which parameter combinations produce interesting outcomes, and which assumptions are load-bearing. The experiments proposed in Chapters 6 and 7 are designed to test whether those assumptions hold in real biology.

Chapter 6

Drug Combinations and Penetration

6.1 Spatial Tumor Simulation: Depth-Dependent Energy Deposition

A drug that works in a petri dish may fail in a patient for reasons that have nothing to do with biology. Tumors are three-dimensional structures with blood vessels at the periphery and drug-starved interiors. Light, ultrasound, and drug molecules each attenuate differently as they travel through tissue. If a treatment can't physically reach cancer cells deep inside a tumor, its biochemical potency is irrelevant.

To address the single-cell model's lack of spatial context, we extended the simulation to a 2D heterogeneous tumor grid (500 x 500 cells, 20 um/cell = 1 cm x 1 cm tissue cross-section) with physics-based energy attenuation. The tumor architecture consists of a glycolytic core (80% glycolytic, 15% OXPHOS, 5% persister), an OXPHOS-rich periphery (70% OXPHOS, 20% glycolytic, 8% persister, 2% persister+NRF2), persister cluster pockets, and a stromal border — totaling 159,033 tumor cells per simulation.

Energy is applied from the tissue surface (row 0) with physically grounded attenuation models:

- **PDT**: Modified Beer-Lambert exponential decay with effective attenuation coefficient 0.31 per mm, yielding a penetration depth of approximately 3.2 mm at 630 nm (Jacques, Phys Med Biol 2013) - **SDT**: Acoustic intensity attenuation at 0.7 dB/cm/MHz and 1 MHz operating frequency (Cobbold, Foundations of Biomedical Ultrasound, 2007) - **RSL3**: Uniform concentration (systemic drug, no depth dependence)

Dead cells release labile iron to their Moore neighborhood (8-neighbors), modeling the bystander ferroptosis effect. Iron is consumed by Fenton chemistry each timestep rather than accumulating permanently.

Results (Figure 8):

Key findings from the spatial simulation:

1. **PDT kill rate collapses beyond 3 mm.** At 1 mm depth, PDT kills 90.8% of

tumor cells; at 5 mm, only 15.7%; at 9 mm, 0.5%. This directly reflects the exponential light attenuation in tissue and quantifies why PDT is limited to superficial tumors.

2. **SDT maintains >88% kill rate through the full 1 cm depth.** Ultrasound attenuation at 1 MHz is mild (∼0.7 dB/cm), preserving sufficient acoustic intensity to activate sonosensitizers at centimeter depths. This validates the paper's central claim: SDT extends ROS-ferroptosis to deep-seated tumors where PDT cannot reach.

3. **RSL3 kills uniformly but weakly (∼3-6%).** As a systemic drug, RSL3 shows no depth dependence — it reaches all cells equally. However, it kills selectively: 42.2% of persisters (4,815/11,391) versus 0% of glycolytic cells, consistent with the single-cell model.

4. **Overall tumor kill: SDT 91.9%, PDT 26.1%, RSL3 3.7%, Control 0.1%.** PDT killed only 26.1% of the heterogeneous tumor because 74% of cells lie beyond its effective penetration depth. SDT killed 91.9% — a 3.5-fold advantage over PDT in this 1 cm tissue model.

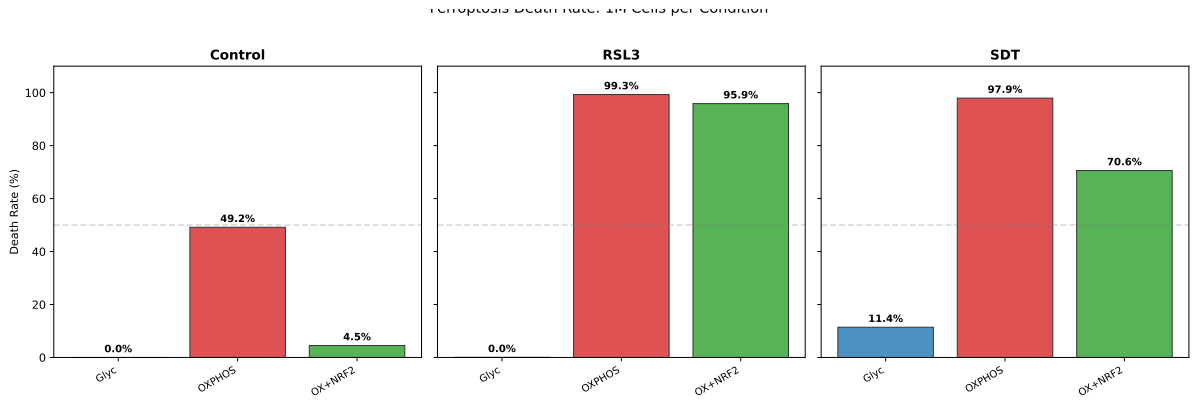

Figure 6.1: Spatial tumor simulation: depth-kill curves and 2D death heatmaps.

The spatial model treats RSL3 as uniformly distributed, but real drug concentration drops with distance from the nearest blood vessel. A tissue-specific drug penetration module using the exponential Krogh cylinder approximation ($C(r) = C_0 \times \exp(-r/\lambda)$, where $\lambda = \sqrt{D/k}$ is the penetration length) shows that even at the vessel wall (the highest-concentration position), tissue-specific vascular permeability reduces effective drug exposure — yielding vessel-wall persister kill rates of 12.1% in well-vascularized epithelial tissue, 2.6% in poorly-vascularized tissue (pancreatic), and 1.8% behind the blood-brain barrier (glioblastoma), with overall tumor kill rates even lower (6.6%, 2.1%, 1.5% respectively). These are substantially lower than the 40% kill rate observed in 2D culture, reinforcing the in-vitro-to-in-vivo translation gap. All transport parameters are estimated, not calibrated — the comparison across tissue types is directionally informative, not quantitatively precise.

What this means. For deep-seated tumors — pancreatic, hepatic, pelvic — any drug that cannot penetrate centimeters into tissue will underperform its in-vitro promise

regardless of its biochemical potency. The in-vitro-to-in-vivo gap is not primarily about biology; it is about physics. This applies to RSL3 and, by extension, to any pharmacologic ferroptosis inducer with similar tissue distribution limitations. SDT maintains efficacy at depth because ultrasound penetrates centimeters with modest attenuation, while light (PDT) is absorbed within millimeters.

How to test this prediction. Grow persister-enriched cancer cells in 3D tumor spheroids (500 μm diameter or larger) and apply PDT, SDT, and RSL3. Measure cell viability at multiple depths within the spheroid using confocal sectioning or optical clearing with viability staining.

Expected if prediction holds: PDT kill exceeds 80% at the spheroid surface but drops sharply toward the core (within 250 μm for a 500 μm spheroid). SDT kill remains high at all measured depths. RSL3 kill drops with distance from the medium supply surface. For larger tissue constructs or in vivo models, the differential should scale with the larger dimensions predicted by the 1 cm spatial simulation.

Expected if prediction fails: All three modalities show similar depth-dependent kill profiles. This would indicate that the model's differential attenuation parameters are wrong, or that drug penetration in 3D culture is better than the Krogh cylinder approximation predicts. Either result would be informative for refining the model.

6.2 Vulnerability Window: How Long Does the Ferroptosis-Sensitive Window Stay Open?

If residual cancer cells are temporarily more vulnerable to ferroptosis after initial treatment, the critical practical question becomes: how long does the window stay open? A vulnerability that exists but closes within hours is biologically real but clinically useless — there is no time to administer a follow-up treatment. A window that lasts weeks gives clinicians a realistic intervention opportunity.

A critical question for clinical translation is timing: how long after chemotherapy withdrawal do persister cells remain ferroptosis-sensitive? We modeled the recovery of defense pathways (FSP1, GPX4, NRF2, GSH) using exponential kinetics with literature-estimated half-times (FSP1: 7 days, GPX4: 3 days, NRF2: 5 days, GSH: 1 day) and simulated 100,000 cells per condition at each timepoint.

The RSL3 vulnerability window closes within 3 days as GPX4 is re-expressed. By one week, RSL3 is completely ineffective ($<$0.01%). **The SDT window remains open for at least 4 weeks** (99.5% kill at day 28), because the exogenous ROS burst overwhelms even partially recovered antioxidant defenses. This differential has direct clinical implications: pharmacologic ferroptosis inducers require intervention within days of chemotherapy, while physical modalities offer a much wider scheduling window. These

window estimates depend on the literature-derived half-life parameters and should be treated as approximate until experimentally validated.

What this means. If the model is correct, the logistics of treatment timing become a critical variable — not just for efficacy, but for determining which modality class is even feasible. A GPX4 inhibitor administered one week after chemotherapy would arrive to find a population of cells that has already re-expressed its primary defense. A physical ROS modality administered at the same timepoint would still find cells whose antioxidant reserves, while partially recovered, remain overwhelmed by the exogenous ROS dose. This converts a biological question (how long are cells vulnerable?) into a practical scheduling question (how quickly can the follow-up treatment be delivered?).

How to test this prediction. After drug withdrawal from a persister-inducing therapy (e.g., osimertinib withdrawal from EGFR-mutant lung cancer cells), measure GPX4, FSP1, NRF2, and GSH levels at serial timepoints (0h, 6h, 1d, 3d, 1wk, 2wk, 4wk). At each timepoint, assess ferroptosis sensitivity by treating matched samples with RSL3 and measuring cell death.

Expected if prediction holds: GPX4 protein levels recover to baseline within 3-5 days (half-life ∼3 days). FSP1 recovers more slowly (half-life ∼7 days). RSL3 sensitivity drops in parallel with GPX4 recovery. Sensitivity to exogenous oxidative stress (e.g., tert-butyl hydroperoxide as a proxy for exogenous ROS, though it does not replicate SDT's spatial or mechanistic features) persists longer because the ROS dose overwhelms partially recovered defenses.

Expected if prediction fails: All defense pathways recover simultaneously rather than sequentially. This would mean the differential half-life estimates in the model are wrong, and there is no pharmacologic-vs-physical window difference — both modality classes face the same timing constraint. Alternatively, recovery may be much faster than modeled (hours, not days), in which case the window is too narrow for any follow-up treatment to exploit.

6.3 Mechanistic Combination Modeling: Why Drug Pairs Synergize

The question behind drug combination design is straightforward: if the cell has two parallel defense mechanisms, what happens when you attack both at once? The ferroptosis pathway has two parallel repair arms: GPX4 (GSH-dependent) and FSP1 (CoQ10-dependent). These feed into a shared antioxidant quench term that suppresses autocatalytic lipid peroxidation propagation. Depleting both arms simultaneously should produce mechanistic synergy — the simulation lets us test this prediction quantitatively.

Running pairwise combinations of four interventions (RSL3/GPX4 inhibitor, SDT/exogenous

ROS, FSP1 inhibitor, HDAC inhibitor/basal ROS upregulation) on 1,000 persister cells per condition and comparing against Bliss-independent predictions reveals three synergistic pairs. RSL3 + FSP1i is the strongest (Bliss synergy score 1.99): single-agent RSL3 kills 40% and FSP1i kills 3.7%, but the combination kills 84.1% — nearly double the 42.2% Bliss prediction. The pathway traces show why: with both GPX4 and FSP1 depleted, mean lipid peroxidation reaches 8.7 (near the 10.0 death threshold), whereas RSL3 alone only drives LP to 4.5 because intact FSP1 partially compensates. RSL3 + HDACi (1.25x) and FSP1i + HDACi (1.21x) are also synergistic. SDT combinations are merely additive because SDT alone already kills 100% of persisters (ceiling effect). Drug potency parameters (FSP1i 85% inhibition, HDACi 2x ROS) are estimates — exact synergy scores depend on these assumptions, but the directional finding (dual-pathway depletion > single-pathway) is robust to parameter variation.

What this means. The 1.99× Bliss synergy is not about the specific drug pair RSL3 + FSP1i. It is about the principle: ferroptosis has two parallel repair pathways, and disabling both is mechanistically more effective than disabling either alone. Any drug combination that simultaneously depletes GPX4-dependent and GPX4-independent repair should produce synergy, regardless of the specific molecules used. This generalizes the prediction beyond the compounds in the simulation — it identifies a strategy (dual-pathway depletion) rather than endorsing a specific regimen. The practical implication for drug development: screen combination strategies by their pathway coverage, not just by their individual potency.

How to test this prediction. Treat persister-enriched cancer cells with a GPX4 inhibitor (RSL3 or ML162) and an FSP1 inhibitor (iFSP1 or DHODH inhibitor brequinar) across a dose matrix. Compute the combination index using Chou-Talalay analysis or compare against Bliss independence predictions. Measure lipid peroxidation (C11-BODIPY) to confirm the mechanistic pathway.

Expected if prediction holds: Combination index < 0.5 (strong synergy) at concentrations where single agents produce 20-50% kill. Lipid peroxidation in the combination group exceeds single-agent levels by at least 50%, consistent with dual-pathway depletion removing the compensation that limits single-agent LP accumulation.

Expected if prediction fails: Combination index > 0.8 (additive or antagonistic). This would indicate that FSP1 does not function as a truly independent parallel repair pathway — perhaps it shares upstream regulators with GPX4, or perhaps the model's independence assumption overstates their biochemical separation. An antagonistic result (CI > 1.0) would be particularly informative, suggesting the two pathways interact in ways the model does not capture.

Chapter 7

Resistance Mechanisms

The simulations in Chapter 5 showed that persister cells are selectively vulnerable to ferroptosis. The spatial and combination models in Chapter 6 showed that this vulnerability is modulated by depth, timing, and pathway targeting. But all of those models assumed a uniform, well-controlled environment — the biological equivalent of a petri dish. Real tumors are nothing like petri dishes.

This chapter examines four features of the tumor microenvironment (TME) that the model predicts will alter treatment outcomes. Three are resistance mechanisms that selectively penalize pharmacologic ferroptosis inducers while leaving physical modalities relatively unaffected: oxygen gradients (Section 7.1), stromal metabolic support (Section 7.3), and acidic pH (Section 7.4). One is an amplification mechanism that favors treatments producing dense spatial kill patterns: immune coupling through DAMP signaling (Section 7.2). Together, they share a structural asymmetry: pharmacologic approaches depend on endogenous cellular processes — basal ROS generation, GSH depletion, drug bioavailability — that the TME can disrupt. Physical modalities deliver exogenous energy that bypasses these dependencies. Whether this asymmetry holds in real tumors is an open question; the experiments proposed in each section below are designed to test it.

7.1 Tumor Microenvironment: Oxygen Gradients and Hypoxia

The center of a solid tumor has almost no oxygen. Blood vessels penetrate only the outer 100-150 micrometers of tissue, and cells deeper than that rely on whatever oxygen diffuses inward — which, for many tumors, is close to zero. This matters for ferroptosis because the earliest step in the cascade — generation of basal reactive oxygen species by the mitochondrial electron transport chain — requires oxygen as the terminal electron acceptor. If there is no oxygen, there is no basal ROS, and a drug like RSL3 that works by disabling the cell's defense against ROS-driven lipid damage has nothing to defend

against. It is like hiring a firefighter in a town where no fires ever start.

The preceding simulations assumed uniform oxygenation — every cell generates the same basal ROS regardless of position. Real tumors have oxygen gradients: O2 penetrates approximately 100-150μm from the vasculature (Vaupel, Cancer Research 1989; modeled via the Krogh cylinder approximation, with underlying O2 transport physiology consistent with Anatomy & Physiology 2e, Ch.22.4), creating hypoxic cores where mitochondrial ETC activity drops and basal ROS production falls (Biology2e Ch.7, Sec.7.4: O2 is the terminal electron acceptor for the electron transport chain).

To quantify this, we applied an exponential O2 gradient from the tumor edge inward (O2(d) = exp(-d/λ), same Krogh approximation used for drug penetration) and compared treatment efficacy under uniform vs gradient conditions. Because the model only modulates basal ROS (not Fenton chemistry or SDT/PDT dose), this is a conservative estimate of hypoxia's protective effect.

The result is striking. Under the O2 gradient ($\lambda = 120\mu$m), RSL3 overall kill collapses from 3.7% to 0.1%, and in the hypoxic zone (deeper than 360μm from the tumor edge) RSL3 kills zero cells. SDT overall kill drops only modestly — from 91.9% to 87.8% — and maintains 86.6% kill even in the hypoxic zone. This differential is robust across all tested penetration lengths (λ = 80, 100, 120, 150μm) and emerges directly from the biochemistry: RSL3 inhibits GPX4 but depends on the cell's own mitochondrial ROS to initiate lipid peroxidation. In hypoxia, there is no basal ROS to drive the cascade, regardless of how thoroughly GPX4 is inhibited. SDT delivers exogenous ROS via sonosensitizer activation (primarily Type I electron-transfer mechanism, less O2-dependent than the Type II singlet-oxygen pathway), bypassing the mitochondrial bottleneck.

This finding has two implications. First, it quantifies a specific mechanism by which the tumor microenvironment protects cells from pharmacologic ferroptosis inducers — not through upregulation of defense pathways (the NRF2/MUFA mechanisms modeled in Chapter 5), but through the absence of the upstream trigger (basal ROS). Second, it provides a mechanistic rationale for why physical ROS modalities might maintain efficacy in contexts where pharmacologic approaches fail: they supply what hypoxic cells lack. This is the same depth-vs-penetration argument made in Section 8.2, but now grounded in TME biology rather than tissue optics alone.

How to test this prediction. Compare RSL3 and SDT efficacy in normoxic (21% O2) versus hypoxic (1% O2) conditions using a hypoxia chamber or 3D tumor spheroids with verified hypoxic cores (pimonidazole staining).

Expected if prediction holds: RSL3 kill drops more than 10-fold in 1% O2 conditions, paralleling the simulation's 3.7% $\rightarrow$ 0.1% collapse. SDT kill drops less than 2-fold (the simulation predicts 91.9% $\rightarrow$ 87.8%). Lipid peroxidation measurements (C11-BODIPY) should show that RSL3-treated cells in hypoxia accumulate minimal LP, confirming that the basal-ROS-dependent cascade cannot initiate without oxygen.

Expected if prediction fails: Both RSL3 and SDT show similar efficacy reductions in hypoxia. This would mean SDT's ROS generation is more O2-dependent than the model assumes — specifically, that the Type I electron-transfer mechanism (which we model as O2-independent) actually requires significant oxygen. Alternatively, RSL3 may maintain efficacy in hypoxia through a mechanism the model does not capture (e.g., direct GPX4 inactivation triggering LP accumulation through non-mitochondrial ROS sources). Either result would revise the model's central assumption about the ROS-source asymmetry.

Caveats: the O2 model treats SDT/PDT as fully O2-independent, which is optimistic for SDT's hypoxic performance (the Type II photodynamic mechanism requires O2 for singlet oxygen generation, and even Type I SDT likely retains some O2 dependence). The Fenton reaction substrate (H2O2) is also O2-derived via the superoxide→SOD pathway, but we modulate only basal_ros, not Fenton rate directly. The O2 field is steady-state (no consumption dynamics) and uses distance from the tumor edge as the O2 source (no explicit vasculature). All of these simplifications mean the actual hypoxic protection is likely STRONGER than what we report.

7.2 Immune Coupling: DAMP-Mediated T Cell Activation

When cancer cells die from ferroptosis, they release molecular alarm signals — damage-associated molecular patterns (DAMPs) — that can activate nearby immune cells. But not all ferroptotic deaths are equal. The amount and spatial pattern of DAMP release depends on how the cell died: a cell overwhelmed by a massive exogenous ROS burst releases more membrane debris and more inflammatory signals than a cell that died from a slow, drug-mediated lipid cascade. If physical modalities produce both more killing and more immunogenic killing, the advantage compounds.

Ferroptotic cell death releases damage-associated molecular patterns (DAMPs) — particularly HMGB1, calreticulin, and ATP — that activate dendritic cells and prime local T cell responses (immunogenic cell death, ICD). To explore whether the kill-density and kill-quality differential between SDT and RSL3 translates into differential immune activation, we added a spatial immune coupling layer to the TME simulation. Dead cells release DAMPs into a diffusion field; local DAMP concentration activates resident T cells via a half-maximal response function (Kd = 5.0); activated T cells kill adjacent tumor cells at rate 0.02 per step, suppressed by a PD-1 brake (70% inhibition baseline, reduced to 14% with simulated anti-PD-1).

The DAMP quality per dead cell is not equal across treatments. Lipid peroxidation is an autocatalytic free-radical chain reaction: each peroxyl radical abstracts a hydrogen from an adjacent polyunsaturated fatty acid, generating a new radical and propagating

the cascade (established biochemistry; for the role of O2 as terminal electron acceptor driving basal ROS via the mitochondrial ETC, see Biology2e Ch.7, Sec.7.4). For high-ROS treatments, this cascade continues post-threshold: the simulation now continues LP accumulation for 5 steps after death (with zero repair — the cell's defenses have failed), producing emergent overshoot. SDT/PDT Persister LP reaches ∼20 (2.0× the death threshold) because exogenous ROS sustains the chain reaction, while RSL3's slower LP accumulation reaches ∼7.8. The DAMP differential therefore comes from both kill *density* (SDT kills ∼140K cells vs RSL3's ∼163) and kill *quality* (each SDT dead cell releases ∼2.6× more DAMPs).

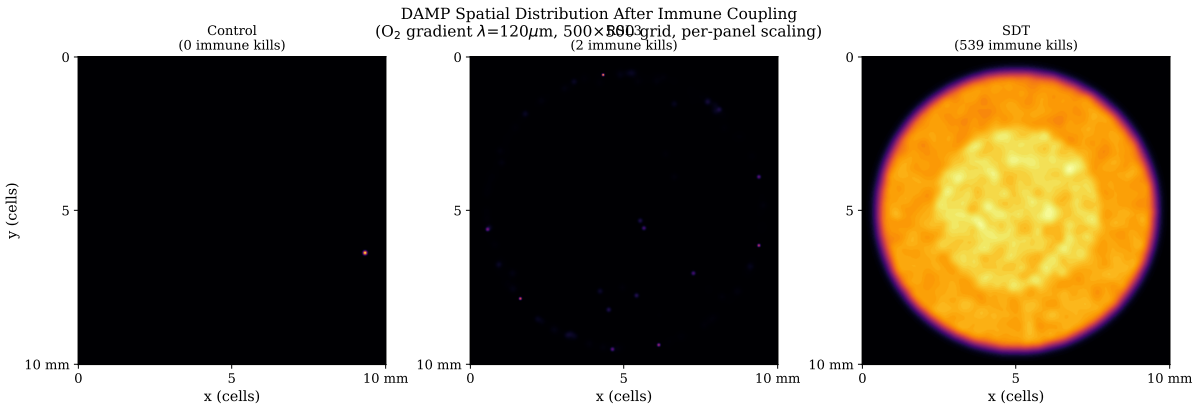

Figure 7.1: DAMP spatial distribution after immune coupling (O_2 gradient λ=120μm, per-panel scaling—intensity not comparable across panels). SDT covers the full tumor area (139,641 kills, 539 immune kills); RSL3 produces sparse isolated hotspots (163 kills, 2 immune kills).

Results (O2 gradient λ=120μm, 500×500 grid): SDT with immune coupling produces 139,640 ferroptotic kills plus 521 immune kills (88.1% total), compared to 87.8% with ferroptosis alone. Adding simulated anti-PD-1 increases immune kills to 1,477 (88.7% total) — a 2.8× amplification. RSL3 produces only 5 immune kills without anti-PD-1 and 18 with it, because its sparse kill field generates negligible local DAMP concentration. The SDT-to-RSL3 immune kill ratio is 104:1 without anti-PD-1 and 82:1 with it.

The absolute immune contribution is modest (∼0.3-0.9% additional kill for SDT), which is expected for a resident T cell model without systemic immune recruitment. The qualitative finding matters more: treatments that produce dense spatial kill patterns generate self-reinforcing immune activation that pharmacologic approaches with sparse kill cannot achieve. This suggests that the physical-modality advantage extends beyond direct ROS delivery — the spatial kill density itself creates a secondary immune mechanism that further widens the gap.

How to test this prediction. Treat persister-enriched cancer cells with SDT versus RSL3 and measure immunogenic cell death (ICD) markers: calreticulin surface exposure (flow cytometry), HMGB1 release (ELISA on culture supernatant), and ATP release

(luminescence assay). For the spatial-density component, compare ICD marker release in 2D monolayer (uniform kill) versus 3D spheroid (spatially structured kill).

Expected if prediction holds: SDT-killed cells release at least 2-fold more HMGB1 and ATP per dead cell than RSL3-killed cells. In 3D spheroids, dense SDT kill regions show measurably higher DAMP concentrations and immune cell recruitment (if co-cultured with dendritic cells or PBMCs) than sparse RSL3 kill regions.

Expected if prediction fails: DAMP release is proportional to the number of dead cells regardless of treatment modality or spatial pattern. This would mean the model's LP-overshoot-driven DAMP quality differential is wrong — immune activation depends on total kill count, not kill geometry or kill quality. The practical implication would be that the physical-modality immune advantage comes from killing more cells (which it does), not from killing them more immunogenically (which would reduce the mechanistic specificity of the prediction).

Caveats: the immune model represents resident T cell activation only (0-48h), not systemic immune recruitment or adaptive immune memory. All immune parameters (kill rate, DAMP diffusion, DC activation threshold) are estimated, not calibrated to experimental data. The LP overshoot is now emergent from the biochemistry (5 post-death steps with zero repair) rather than an estimated multiplier. PD-1/anti-PD-1 parameters are simplified (single brake coefficient). These results establish qualitative directionality — dense ferroptotic kill enables immune amplification — but the absolute magnitudes should not be treated as predictions.

7.3 Stromal Shielding: CAF-Mediated Protection at the Tumor Periphery

The preceding TME features modeled the tumor interior (hypoxia, immune coupling). But the tumor boundary presents a different biology: cancer-associated fibroblasts (CAFs) at the tumor-stroma interface supply neighboring tumor cells with cysteine (for GSH synthesis via the GGT1/SLC7A11 pathway) and oleic acid (MUFA for membrane remodeling via ACSL3-mediated uptake). This metabolic cooperation has been documented in CAF co-culture studies **?** and lipid-ferroptosis research **?**; **?**, but has not been quantified in a spatial ferroptosis simulation.

To test whether stromal shielding meaningfully affects treatment outcomes, we identified the 1,796 tumor cells (1.1% of 159,033 total) with at least one stromal Moore neighbor and applied per-step CAF-mediated boosts: GSH +0.06/step (2.4× the endogenous NRF2 recovery rate, capped at 18.0 mM) and MUFA +0.003/step (capped at 0.25). These are estimated parameters — no textbook coverage exists for CAF biology in any of the reference textbooks used in this project. The underlying biochemistry (GSH scav-

enging, MUFA-mediated PUFA displacement) is handled by the existing ferroptosis-core engine; the CAF contribution simply increases the supply of two defensive metabolites to boundary cells.

The result is treatment-selective. Among the 1,796 stromal-adjacent cells, CAF protection halves the RSL3 kill rate from 3.0% to 1.5% (50% relative reduction) but reduces the SDT kill rate from only 96.1% to 91.2% (5.1% relative). The hypoxic zone kill rate is unchanged for all treatments — CAFs are at the boundary, not the interior. The overall kill rate change is small (∼0.06% for SDT, ∼0.02% for RSL3) because only 1.1% of tumor cells are affected.

The mechanism behind the treatment selectivity is the same ROS imbalance that drives the hypoxia finding: RSL3 depends on slow basal ROS (∼0.3 relative units) to drive the LP cascade through compromised GPX4. A 2.4× GSH supply boost meaningfully slows this cascade — each step, more ROS is scavenged before it reaches lipid membranes. SDT delivers ∼17× higher exogenous ROS (∼5.0 relative units), overwhelming the CAF defense within a few steps regardless of the GSH boost. The Michaelis-Menten scavenging kinetics (GSH of 18 provides only 26% better scavenging than GSH of 5) ensure that even maximal CAF supply cannot rescue cells from SDT's ROS flood.

This finding complements the hypoxia and immune coupling results: physical modalities bypass both hypoxic protection (Section 7.1) and CAF-mediated stromal shielding, while also generating stronger downstream immune activation (Section 7.2). Hypoxia and stromal shielding are direct resistance mechanisms that blunt pharmacologic ferroptosis induction; immune coupling is a separate amplification effect that favors treatments with dense spatial kill patterns.

How to test this prediction. Co-culture persister-enriched cancer cells with cancer-associated fibroblasts (CAFs) isolated from patient tumors or established CAF cell lines. Compare ferroptosis markers (C11-BODIPY lipid peroxidation, GSH quantification, cell viability) in four conditions: tumor cells alone + RSL3, tumor cells alone + SDT, tumor cells with CAFs + RSL3, tumor cells with CAFs + SDT.

Expected if prediction holds: CAF co-culture reduces RSL3-induced cell death by more than 30% (the simulation predicts 50% relative reduction) while reducing SDT-induced death by less than 10% (simulation: 5.1% relative). GSH levels in tumor cells adjacent to CAFs should be measurably higher than in tumor cells without CAF contact, confirming the metabolic supply mechanism.

Expected if prediction fails: CAFs protect equally against both RSL3 and SDT. This would indicate either (a) CAFs employ defense mechanisms beyond GSH and MUFA supply that also attenuate exogenous ROS, or (b) the model's ROS-dose asymmetry (0.3 basal vs 5.0 exogenous) overstates the real difference, and CAF-supplied antioxidants CAN neutralize SDT's ROS burst. A finding that CAFs provide NO protection against either treatment would suggest the entire metabolic-cooperation model is weaker than

the literature implies.

Caveats: all stromal parameters are estimated, not calibrated to experimental data. The model treats all boundary stromal cells as equally protective CAFs (in reality, CAF density and phenotype vary). CAF metabolic exhaustion is not modeled — the supply rate is constant regardless of demand. ECM-mediated drug diffusion barriers are not included (deferred to a future iteration). Only the outermost ring of tumor cells (depth = 1 cell layer) receives CAF protection; paracrine signaling range may extend further. These results establish that stromal shielding is treatment-selective in the model, but the absolute magnitudes depend on estimated parameters.

7.4 Acidic pH: Iron Release vs Drug Ion Trapping

Tumor cores are acidic (pH 6.5-6.8) due to lactic acid accumulation from aerobic glycolysis (the Warburg effect — not covered in the reference textbooks, which associate lactic acid fermentation with anaerobic conditions only). This acidity creates two competing effects on ferroptosis: (1) ferritin protein shells destabilize below pH ∼6.5, releasing stored iron into the labile iron pool and increasing Fenton-driven ROS generation (pro-ferroptosis); (2) weak-base drugs become protonated at low extracellular pH and are trapped in acidic compartments, reducing intracellular drug availability (anti-ferroptosis, governed by Henderson-Hasselbalch equilibria — Chemistry2e Sec.14.2). The net effect on ferroptosis is not obvious from first principles: does the iron boost or the drug loss dominate?

To quantify this, we applied an exponential pH gradient from the tumor edge inward ($pH(d) = 7.4 - 0.9 \times (1 - \exp(-d/\lambda))$, λ=120μm), yielding pH 7.4 at the well-perfused periphery and ∼6.5 in the deep core. The iron modulation (2.35× at pH 6.5) increases the Fenton ROS term, while the ion trapping (36% drug loss at pH 6.5, linearized Henderson-Hasselbalch) reduces RSL3's effective GPX4 inhibition from 92% to 59%.

In the model, drug trapping dominates. RSL3 ferroptosis kills drop from 163 to 77 (53% reduction) because the GPX4 restoration (from 8% to 41% of normal activity) is a 5.1× repair boost, while the iron-driven ROS increase is only 7%. SDT is barely affected (+0.8%, from 139,640 to 140,693 ferroptosis kills) because ultrasound energy delivery is pH-independent — there is no drug to trap — and the iron boost adds only 0.4% to SDT's total ROS (dwarfed by exogenous ROS of 5.0 relative units). Control shows a tiny increase (1 to 2 ferroptosis kills) from the Fenton boost alone.

This adds a third direct resistance mechanism — alongside hypoxia (Section 7.1) and stromal shielding (Section 7.3) — that selectively penalizes pharmacologic ferroptosis inducers while leaving physical modalities unaffected. All three share the same structural asymmetry: pharmacologic approaches depend on endogenous processes that the TME can disrupt — basal ROS generation (removed by hypoxia), GSH depletion (counteracted

by stromal cysteine supply), and intracellular drug bioavailability (reduced by pH-driven ion trapping) — while physical modalities deliver exogenous energy that bypasses these vulnerabilities.

How to test this prediction. Measure RSL3 efficacy at pH 7.4 versus pH 6.5 in buffered culture medium. Additionally, measure intracellular RSL3 concentration (or a fluorescent RSL3 analog) at both pH values using HPLC or fluorescence microscopy to directly test the ion-trapping mechanism.

Expected if prediction holds: RSL3-induced cell death drops by more than 40% at pH 6.5 compared to pH 7.4. Intracellular RSL3 concentration at pH 6.5 is measurably lower than at pH 7.4, consistent with extracellular protonation and trapping. SDT efficacy is pH-independent (no drug to trap).

Expected if prediction fails: RSL3 efficacy is pH-independent. This would be a significant finding: it would mean that either (a) RSL3 is not subject to Henderson-Hasselbalch ion trapping because its chloroacetamide chemistry behaves differently from classic weak-base drugs (the most likely explanation for failure, since RSL3's pKa is poorly characterized), or (b) the iron release at low pH compensates for any drug trapping, producing a net-neutral effect. Either result would narrow the model's predictions substantially — the pH resistance mechanism would be invalidated, leaving only hypoxia and stromal shielding as TME resistance factors.

This is the finding most in need of direct experimental validation, because the ion_trap_sensitivity parameter is the most uncertain value in the model. RSL3 is a chloroacetamide, not a classic weak base, and its behavior at acidic pH has not been empirically characterized.

Caveats: all pH parameters are estimated. RSL3 is a chloroacetamide, not a classic weak base — its pKa and ion trapping behavior are poorly characterized, making the ion_trap_sensitivity the most uncertain parameter in this model. The linear Henderson-Hasselbalch approximation is valid over the narrow pH 6.5-7.4 range but would break down at more extreme pH values. The pH field is steady-state (no lactate production or buffering dynamics). Tumor pH ranges (6.5-6.8) are from primary literature; no reference textbook covers tumor-specific pH data.

7.5 The Cumulative Picture

The four TME features explored in this chapter are not independent effects that operate in isolation. In a real solid tumor, a cancer cell deep in the hypoxic core (Section 7.1) sits in an acidic microenvironment (Section 7.4), may receive metabolic support from nearby stromal cells if it is close to the tumor boundary (Section 7.3), and — if killed — contributes to a DAMP field that either does or does not activate local immune cells depending on the spatial density of surrounding cell death (Section 7.2).

The three resistance mechanisms share a structural asymmetry: pharmacologic ferrop-

tosis inducers depend on endogenous cellular processes — basal ROS generation (removed by hypoxia), GSH homeostasis (supplemented by stromal supply), and intracellular drug concentration (reduced by pH-driven ion trapping) — while physical modalities deliver exogenous energy that bypasses all three dependencies. The immune coupling mechanism compounds this by favoring the dense, high-quality kill patterns that physical modalities produce.

Whether this asymmetry translates to real tumors is an empirical question. The experimental proposals in each section above are designed to test the individual mechanisms. If any one mechanism fails to replicate, the others remain independently testable — they share a common theme but are mechanistically distinct. If all three resistance mechanisms replicate, the cumulative effect in a real tumor would be substantial: pharmacologic ferroptosis inducers would face hypoxic protection in the core, stromal protection at the boundary, and chemical trapping throughout — while physical modalities would bypass all three barriers simultaneously.

The honest assessment: this is a hypothesis supported by computational modeling with estimated parameters, not a demonstrated clinical advantage. The experiments proposed above are designed to distinguish between "the model identified a real biological pattern" and "the model produced an artifact of its simplifying assumptions." Both outcomes advance understanding.

Chapter 8

Counterarguments and Failure Modes

8.1 Other Directions The Repo Was Underweighting

One reason to widen the article framing is that the repo is no longer only telling a PDT/SDT story. Even under conservative tagging, several adjacent directions deserve explicit mention because they may compete with or complement the physical-ROS hypothesis.

Radioligand therapy is the clearest example of a clinically advancing, spatially selective modality-class competitor. It shares the intuitive appeal of localized tumor killing while operating within a much more mature theranostic and dosimetry framework.

Immune residual-disease strategies — especially TIL/TCR-style cell therapy, therapeutic mRNA or neoantigen vaccination, and phagocytosis-checkpoint manipulation — are important because they aim not just to kill a vulnerable state, but to convert residual disease into an immune-recognizable target. The revised mRNA-vaccine tagging pass is a concrete reminder that this lane must be interpreted through therapeutic-cancer-vaccine papers rather than the broader post-COVID vaccine literature.

Pathway-target layers such as SCD/MUFA remodeling, DHODH-linked backup defenses, DHCR7/7-DHC biology, and TRIM25/GPX4 degradation matter because they may explain why ferroptosis-looking vulnerabilities repeatedly weaken in vivo.

Stromal shielding, senescence programs, and broader tumor-microenvironment biology are not side notes. They may be the main reason apparently elegant modality-level ideas stall during translation.

The practical implication is that resistant-state matching should become the primary decision layer. Modality choice matters, but it should come after the biology of residual survival is defined. These alternative directions are not merely interesting — some represent clinically established competitors (radioligand therapy, checkpoint immunotherapy), others represent biological realities that constrain feasibility (stromal shielding, lipid remodeling), and all represent the landscape against which any new hypothesis must justify

its additional complexity and risk.

8.2 The Modality Question: PDT, SDT, and the Tissue-Penetration Tradeoff

If OXPHOS-resistant persister cells are ferroptosis-vulnerable, which modality best exploits this? Our initial corpus analysis identified SDT as unusually aligned with ferroptosis among physical ablative modalities. However, broader PubMed verification reveals a critical context we must address honestly:

Table 8.1: Ferroptosis engagement across physical modalities (PubMed, March 2026).

Modality	Ferroptosis	Ferro+ICD	Depth
PDT	**355**	**67**	Superficial (mm)
SDT	**121**	**25**	Deep (cm)
IRE	15	emerging	Invasive
HIFU	3	minimal	Deep (cm)
TTFields	0	0	Surface

PDT dominates SDT on every ferroptosis and ICD metric by approximately 3:1. This fundamentally challenges any claim that SDT is uniquely positioned for ferroptosis-ICD. PDT — using light rather than ultrasound to activate sensitizers — triggers the same ROS → GSH depletion → ferroptosis → ICD chain with a larger evidence base, more clinical experience, and approved photosensitizers.

SDT's remaining advantage is tissue penetration. Ultrasound penetrates centimeters into tissue; light penetrates millimeters. For deep-seated tumors (pancreatic, hepatic, pelvic), SDT can reach targets that PDT cannot. For accessible superficial tumors (skin, oral, esophageal, bladder surface), PDT is the established modality and SDT offers no advantage.

The honest framing is therefore: SDT is not the unique ferroptosis-ICD modality — PDT is the established one. SDT is a **depth-extended variant** of the same photosensitizer-based ROS-ferroptosis-ICD approach, using ultrasound instead of light to reach tumors that PDT cannot access. This is a meaningful clinical advantage for deep tumors, but it is incremental, not paradigmatic.

The mechanism is shared: sensitizer-activated ROS → GSH depletion → GPX4 inactivation → ferroptosis → DAMP release → ICD Y et al. [2021a]; Q et al. [2021]; Y et al. [2021b]; J et al. [2021]. SDT-specific data shows 72 GSH/GPX4 articles (11.7% of SDT-tagged articles) and 6 dual-pathway articles J et al. [2021]; R et al. [2024]; H et al. [2025b]. A recent Nature Communications paper demonstrates that IRE can also be

combined with ferroptosis via iron-based nanoparticles L et al. [2026], further indicating that the ferroptosis-physical connection is broader than SDT alone.

Critical caveats remain: Most SDT-ferroptosis data involves engineered nanosonosensitizers Q et al. [2021]; Y et al. [2021b]. No clinical study has verified SDT-induced ferroptosis or ICD in human tumors. The PDT precedent — 40 years of development without demonstrating robust ICD-mediated immune synergy in randomized trials — applies as a direct cautionary parallel.

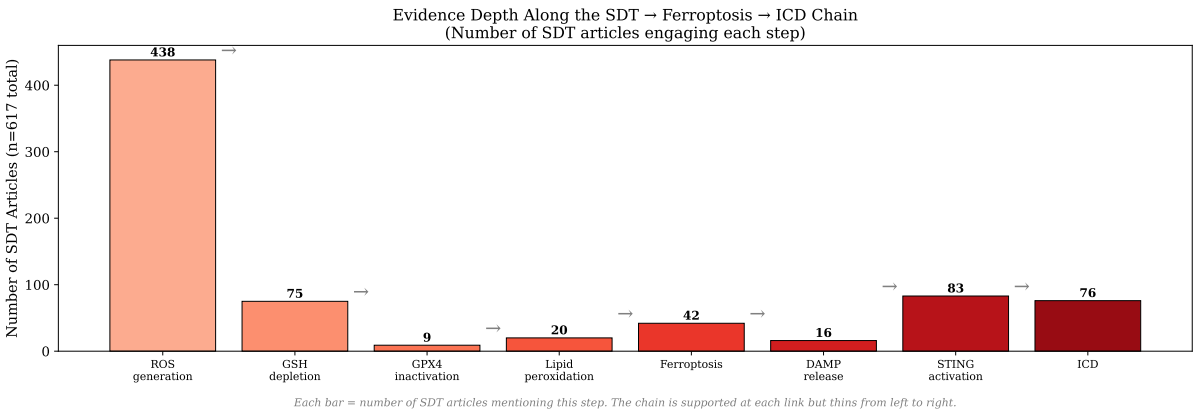

Figure 8.1: SDT ferroptosis-ICD chain evidence.

8.3 Counterarguments, Precedents, and Reasons This Could Fail

The following counterarguments address whether the physical-ROS hypothesis holds biologically and clinically — whether the real-world biology supports the idea, independent of the computational model that generated it.

The photodynamic therapy precedent. PDT is conceptually identical to SDT: physical energy (light vs ultrasound) activates a sensitizer to generate ROS, causing cell death with ICD potential. PDT has been in clinical development for 40+ years and has achieved approval for a few indications (esophageal, skin) but has never demonstrated the "ROS → ICD → immune synergy → durable response" chain in a randomized clinical trial. PDT + immunotherapy combinations remain at the case report level Y et al. [2026]. If this logic worked robustly, PDT would have shown it by now. SDT's claimed advantages over PDT — deeper tissue penetration and potentially superior ferroptosis engagement via engineered nanosonosensitizers — are plausible but incremental, not paradigmatic. The SDT hypothesis must explain why it would succeed where PDT has not demonstrated clear immune synergy.

The ferroptosis vulnerability window may be transient. Persister cells downregulate FSP1 as part of the mesenchymal/dedifferentiated state M et al. [2026]. However,

persisters are a transitional population: they eventually either die or re-enter proliferation with restored drug resistance. The ferroptosis-sensitive window may last days to weeks. If the real window closes before any follow-up treatment can be administered, the therapeutic opportunity is lost regardless of which modality is used.

The lipid-remodeling and backup-defense problem. OXPHOS-resistant cells may co-upregulate NRF2-driven antioxidant defenses, SCD/MUFA remodeling, DHODH-linked backup defenses, DHCR7/7-DHC biology, TRIM25-linked GPX4 degradation dynamics, and other lipid-handling programs that neutralize ferroptosis vulnerability even as the metabolic preconditions accumulate. The pathway-target layer now shows that these escape axes are present in the corpus, but mostly in review-heavy or early mechanistic form. They therefore strengthen the need for caution more than they strengthen any single therapeutic claim.

The nanosonosensitizer confound. Most SDT-ferroptosis papers use engineered nanoparticles *designed* to deplete GSH and release iron. If plain SDT with conventional sonosensitizers doesn't trigger ferroptosis, the insight is about nanoparticle design, not SDT as a modality — and the regulatory path becomes device+drug+nanoparticle, the most complex category.

The ICD-to-immunity chain has repeatedly underperformed clinically. PexaVec (oncolytic virus ICD, failed Phase 3) M et al. [2019], CheckMate-498 (immunotherapy in GBM, failed) A et al. [2023], KEYNOTE-361 (ICI+chemo in urothelial, failed) T et al. [2021], and BIND-014 (nanoparticle, failed Phase 2) Von Hoff et al. [2018] all demonstrate that preclinical rationale frequently fails in translation. SDT has no reason to be exempt from this base rate.

Narrow applicability. The physical-ROS hypothesis applies only to accessible or image-localizable solid tumors, immunocompetent settings where ICD could matter, tumors that actually enter a ferroptosis-prone residual state, and a specific post-therapy time window. Hematologic malignancies, diffuse metastatic disease, and many non-localizable residual states fall outside its plausible range.

Evolutionary limitation. This hypothesis addresses one resistance transition. After SDT kills ferroptosis-sensitive persisters, survivors are selected for ferroptosis resistance (GPX4 upregulation, NRF2 activation, FSP1 restoration). The approach adds one step to the therapeutic arms race but does not end it. Durable eradication would require the ICD-triggered immune response to eliminate residual heterogeneous disease — an outcome that has not been demonstrated for any ICD-based strategy in randomized trials.

Prior art. Collateral sensitivity (Pluchino et al., 2012), adaptive therapy (Gatenby et al., 2009), and synthetic lethality are established frameworks for exploiting resistance tradeoffs. What our analysis adds is the specific SDT connection, not the general principle.

8.4 Structural Limitations of the Simulation Architecture

The counterarguments above address whether the physical-ROS hypothesis holds biologically and clinically. A separate question is whether the computational model that generated the hypothesis is structurally adequate to support it. Section 5.4 listed what the model omits. Here we assess how those omissions affect the specific predictions in Chapters 6 and 7.

Steady-state environmental fields. The oxygen, pH, and drug concentration fields in the TME simulations are computed once based on distance from the tumor edge. They do not change as cells consume oxygen, produce lactate, or metabolize drugs. In real tumors, blood flow fluctuates on timescales of minutes to hours, and cells in the transition zone between oxygenated and hypoxic regions experience intermittent hypoxia — cycling between states rather than remaining permanently in one. Intermittent hypoxia is documented to produce biological responses distinct from chronic hypoxia, including increased angiogenesis, enhanced metastatic potential, and treatment resistance through mechanisms that differ from those modeled here. Impact on findings: the hypoxia result (Section 7.1, RSL3 collapses from 3.7% to 0.1%) would still hold for the permanently hypoxic core — cells deep in the tumor are chronically oxygen-starved regardless of flow fluctuations. But the transition zone, where the model represents a smooth exponential gradient, would in reality be noisier, with cells experiencing fluctuating conditions that the steady-state model cannot capture. The pH result (Section 7.4) faces the same limitation: steady-state pH does not capture lactate production and buffering dynamics. Overall direction: the resistance mechanisms identified here likely persist, but their spatial boundaries would be less sharp than the smooth gradients imply, and the transition zone may harbor a population of cells whose behavior the model cannot predict.

Simplified immune layer. The immune coupling model (Section 7.2) represents resident T cell activation over a 0-48 hour window. It does not model systemic immune recruitment from blood or lymph nodes, adaptive immune memory formation, immune editing of tumor populations, or regulatory immune suppression. The 104:1 SDT-to-RSL3 immune kill ratio is a local phenomenon driven by local DAMP concentration. Whether these local patterns translate to systemic anti-tumor immunity — the outcome that would matter clinically — depends on processes the model does not represent. Two opposing outcomes are plausible. If local DAMP patterns recruit circulating immune effectors, the SDT advantage would compound. If systemic immune regulation dominates — through regulatory T cells, myeloid-derived suppressor cells, or checkpoint upregulation — the local response would be dampened regardless of DAMP density, and the 104:1 ratio would overstate the real differential. Given the clinical trial failure rate documented in Section 8.3 — where ICD-to-immunity chains have repeatedly underperformed across multiple

modalities — the dampening scenario is the more probable default. The amplification scenario would require SDT to produce qualitatively different immune activation than the modalities that have already failed, a claim for which no evidence currently exists. The experimental test proposed in Section 7.2 (ICD marker measurement with immune cell co-culture) is designed to provide the first empirical constraint, but resolving the systemic question will require in vivo models that the current computational framework cannot replace.

No adaptive evolution. The simulation runs 180 steps within a single treatment window. It does not model mutation, clonal selection, or population dynamics across treatment cycles. Section 8.3 already notes the biological reality: survivors of ferroptosis-inducing treatment will be selected for ferroptosis resistance (GPX4 upregulation, NRF2 activation, FSP1 restoration). The computational limitation adds a further constraint: the model structurally cannot quantify how fast this erosion occurs or whether repeated treatment cycles produce accumulating benefit or diminishing returns. The treatment advantages shown in Chapters 6 and 7 — hypoxic resistance bypass, spatial kill density, dual-pathway synergy — are snapshots of what happens during one treatment cycle. Whether those per-cycle advantages translate to durable clinical benefit depends on the rate of evolutionary escape relative to the rate of immune clearance, a question that requires population-dynamics modeling across weeks to months. Such modeling does not yet exist for the ferroptosis context, and building it is a separate research challenge.

Estimated parameters and the quantitative-versus-directional distinction. Many model parameters — CAF supply rates, DAMP diffusion coefficients, T cell kill rates, FSP1 repair kinetics, ion trapping sensitivity — are estimated from indirect evidence rather than calibrated to direct experimental measurements. The sensitivity analysis in Chapter 5 tested robustness to ±50% perturbation on 11 rate constants and found that the qualitative result (Persister cells are more sensitive than Glycolytic cells under SDT) held in 22 of 22 conditions, indicating directional robustness. But directional robustness is not quantitative accuracy. The specific numbers reported throughout this book — 1.99× Bliss synergy (Section 6.3), 53% pH-driven RSL3 kill reduction (Section 7.4), 104:1 immune kill ratio (Section 7.2), 50% stromal protection of RSL3 (Section 7.3) — depend on parameter values that could change substantially with better data. Different FSP1 inhibitor potency would change the synergy score. Different RSL3 pKa would change the pH finding entirely — Section 7.4 explicitly flags this as the most uncertain parameter in the model. The distinction the reader should draw: directional findings (dual-pathway depletion outperforms single-pathway; physical ROS bypasses TME resistance mechanisms; dense kill patterns generate more immune activation than sparse patterns) are robust to parameter variation and are the predictions most worth testing. Specific magnitudes are not robust and should be treated as order-of-magnitude estimates, not precise forecasts. Each experiment proposed in Chapters 6 and 7 is designed

to provide calibration data the model currently lacks — turning estimated parameters into measured ones, and testing whether the directional predictions survive contact with real biology.

These structural limitations define the boundaries of what the computational work can claim. Whether the directional findings survive contact with real biology is the open question that this book cannot answer computationally. The experiments proposed in Chapters 6 and 7 would begin to answer it — but none have been performed, and each could fail for reasons the model does not anticipate.

Part IV

What Should Happen Next

Chapter 9

Research Roadmap

9.1 From Predictions to Experiments

Chapter 8 catalogued the reasons this hypothesis could fail — biologically, clinically, and computationally. This chapter proposes a research program designed to test whether it does. The program is organized not by which finding seems most exciting but by which predictions rest on the best-characterized parameters, because the failure of a high-confidence prediction is more informative than the failure of one that depends on uncertain estimates.

Chapters 6 and 7 each proposed specific experiments attached to individual simulation findings — seven in total, each with a hypothesis, model system, and expected outcomes if the prediction holds or fails. This chapter does not restate those proposals. Instead, it organizes them into a phased program: which experiments to run first and why, which experiments gate which follow-ups, what new experiments are needed beyond the individual-finding tests, and what the field learns if the entire program fails.

This chapter is for research prioritization, not treatment advice. The experimental suggestions identify what to test and why — they are not clinical protocols, statistical designs, or grant-ready specifications. A researcher using this chapter as a starting point would still need to optimize assay conditions, determine sample sizes, and adapt the proposals to their specific model systems and expertise.

9.2 Confidence Tiers

Not all simulation predictions are equally trustworthy. Some rest on well-characterized biochemistry and measured physical constants; others depend on parameters that are estimated from indirect evidence or, in one case, entirely uncharacterized. Assigning a confidence tier to each prediction makes the program's risk profile explicit and helps a researcher decide where to invest effort first.

The tiers are defined by parameter provenance:

- **High confidence**: The prediction depends primarily on textbook biochemistry or well-measured physical constants. Sensitivity analysis confirms directional robustness. Failure of a high-confidence prediction would require a fundamental assumption to be wrong. - **Medium confidence**: The underlying mechanism is published, but key magnitude parameters are estimated from indirect evidence rather than measured in the model system. The prediction's direction is robust but its quantitative magnitude could shift substantially with better data. - **Low confidence**: The prediction depends on parameters that are explicitly flagged as the model's most uncertain. The prediction could change qualitatively — not just quantitatively — if the parameter is wrong.

High confidence:

Hypoxia resistance (Section 7.1). The model predicts RSL3 kill collapses more than 30-fold in hypoxic conditions while SDT maintains greater than 85% kill. This rests on established biochemistry: oxygen is the terminal electron acceptor for mitochondrial ROS generation (textbook), and SDT's Type I mechanism is documented as less O2-dependent than Type II. If this prediction fails, the endogenous-vs-exogenous ROS asymmetry — the most fundamental claim in the simulation work — would need revision.

Spatial penetration differential (Section 6.1). PDT kill collapses beyond 3 mm depth; SDT maintains efficacy through 1 cm. This is physics, not biology: light attenuation at 630 nm and acoustic attenuation at 1 MHz are well-measured constants. If this prediction fails in a spheroid experiment, the attenuation model is wrong — which would be surprising for well-characterized physics but would be immediately informative.

Dual-pathway synergy (Section 6.3). RSL3 + FSP1 inhibitor produces approximately 2x Bliss synergy through dual-pathway depletion. GPX4 and FSP1 independence is published (Bersuker et al., Nature 2019). The directional prediction (dual > single) is robust to parameter variation. If the combination index shows additivity rather than synergy, FSP1's role as an independent parallel repair pathway needs reassessment — a finding with implications well beyond this project.

Medium confidence:

Vulnerability window timing (Section 6.2). RSL3 sensitivity window closes within 3 days; exogenous-ROS sensitivity persists for weeks. The defense-pathway recovery half-lives (GPX4: 3 days, FSP1: 7 days) are literature-estimated, not measured in the specific persister model system the simulation uses. If recovery is much faster than estimated (hours instead of days), the therapeutic window may be too narrow for any follow-up treatment to exploit.

Immune coupling (Section 7.2). SDT generates 104x more immune kills than RSL3 through spatial DAMP density. The LP overshoot that drives DAMP quality is emergent from the biochemistry, but all immune parameters (T cell kill rate, DAMP diffusion, DC activation threshold) are estimated. Chapter 8.4 flags that systemic immune dynamics

could amplify or suppress the local advantage — and the clinical trial failure rate (Section 8.3) suggests suppression is the more probable default.

Stromal shielding (Section 7.3). CAF-mediated GSH and MUFA supply halves RSL3 kill at the tumor boundary while barely affecting SDT. The CAF supply mechanism is published, but the specific supply rates used in the model have no textbook source. The overall tumor impact is small (1.1% of cells affected) even if the per-cell effect is large.

Low confidence:

pH ion trapping (Section 7.4). Acidic pH reduces RSL3 kill by 53% through Henderson-Hasselbalch-driven drug trapping. RSL3 is a chloroacetamide, not a classic weak base — its pKa is uncharacterized, and the ion trapping sensitivity is the most uncertain parameter in the entire model. This prediction is the most likely to fail experimentally. If it does fail, the pH resistance mechanism is eliminated, leaving only hypoxia and stromal shielding as TME resistance factors. If it succeeds, it identifies a previously unrecognized pharmacokinetic limitation of chloroacetamide-class compounds in acidic tumors.

9.3 Program Structure

The experimental program has a gated structure designed to maximize information per experiment and minimize wasted effort.

Phase 1 — High-confidence in-vitro tests. Run the hypoxia (7.1), spatial penetration (6.1), and dual-pathway synergy (6.3) experiments first. These test the cleanest predictions using the best-characterized parameters. If any of these fails, the specific model assumption that broke should be identified before proceeding — it may require revising the simulation architecture before the remaining predictions are worth testing.

Phase 2 — Medium-confidence refinements. If Phase 1 confirms the directional findings, run the vulnerability window (6.2), immune coupling (7.2), and stromal shielding (7.3) experiments. These refine the picture: how long is the treatment window? Does kill density actually translate to immune activation? Does stromal protection matter at the population level? Phase 2 experiments depend on the directional findings from Phase 1 holding — if drugs and physical modalities show similar hypoxic sensitivity, the TME-resistance framing needs revision before investing in refinement experiments.

Phase 3 — Low-confidence test. Run the pH ion-trapping experiment (7.4) last among the in-vitro studies. This is the most likely to fail and the result either validates or eliminates one of the three proposed TME resistance mechanisms. Running it after Phases 1-2 means the broader program isn't derailed by a single uncertain parameter.

Running in parallel — Broader comparison (Section 9.4). This is analytical, not experimental. It runs alongside the wet-lab program and asks whether the entire approach justifies its complexity relative to existing alternatives.

Gated — In-vivo integration (Section 9.5). This is the definitive test and the

most expensive. It should only proceed after the in-vitro program confirms the directional predictions. Running it prematurely would risk investing substantial resources in an approach whose foundational assumptions haven't been verified.

9.4 Broader Comparison: Does This Hypothesis Justify Its Complexity?

This is not an experiment. It is an analytical evaluation of whether the physical-ROS ferroptosis approach is worth pursuing relative to existing alternatives — a question the simulation cannot answer because it doesn't model the alternatives.

The comparison should evaluate each approach on shared axes: regulatory maturity (how far from clinical use?), evidence depth (clinical trials vs preclinical vs computational?), delivery burden (how complex is the treatment logistics?), and applicability to residual disease (does it target the residual-cell biology this project focuses on?).

The most informative comparison is against radioligand therapy. Radioligand uses targeted radiation delivered by molecules that seek specific receptors on cancer cells — it shares the intuitive appeal of spatially selective tumor killing while operating within a more mature regulatory framework (FDA approvals for prostate and neuroendocrine cancers, established dosimetry pipelines, and a large clinical trial base). If radioligand therapy addresses the same clinical need — treating localized residual disease — with a simpler development path, the physical-ROS hypothesis becomes academically interesting but clinically unnecessary.

The comparison is not winner-take-all. Different approaches may be optimal for different clinical contexts: radioligand for tumors with targetable receptors, checkpoint combinations for immunologically hot tumors, and physical-ROS modalities for deep-seated tumors without targetable receptors in immunocompetent settings where ICD might matter. The value of the comparison is not in crowning a winner but in defining the niche — if any — where the physical-ROS approach adds something the alternatives cannot.

9.5 In-Vivo Integration

The in-vitro experiments in Phases 1-3 test individual predictions in controlled conditions. The in-vivo integration experiment tests whether those predictions survive the full complexity of a living system — where hypoxia, pH gradients, stromal shielding, immune activation, drug penetration, and evolutionary dynamics all operate simultaneously.

The experiment should use a syngeneic mouse model (immunocompetent, so the immune coupling prediction can be tested) with a persister-generating protocol (chemotherapy followed by recovery period to select residual disease). Treatment arms must include

honest comparators: SDT with anti-PD-1 (the strongest prediction from the simulation), PDT with anti-PD-1 (the established-modality control), RSL3 alone (the pharmacologic ferroptosis arm), and RSL3 plus FSP1 inhibitor (the dual-pathway combination). Primary endpoint: progression-free survival. Secondary endpoints: immune infiltrate composition, ferroptosis markers in treated tumors, and lipid peroxidation at tumor core versus periphery.

This experiment is expensive, time-consuming, and tests multiple predictions simultaneously. Running it without in-vitro confirmation from Phases 1-2 would be premature because failure would be uninterpretable — you would not know which prediction failed or why. Running it after in-vitro confirmation means failure would identify the specific gap between controlled conditions and biological reality, which is itself valuable.

If the predictions hold: SDT plus anti-PD-1 outperforms other arms in deep tumors with hypoxic cores, with higher immune infiltrate and ferroptosis markers at depth than PDT or RSL3 arms. If the predictions fail: the TME resistance asymmetry does not translate in vivo, which could mean that (a) the in-vitro-to-in-vivo gap is larger than the spatial simulation predicts, (b) the immune coupling does not scale from local DAMP signaling to systemic anti-tumor immunity, or (c) evolutionary escape operates faster than the one-cycle snapshot suggests.

9.6 What the Field Learns If the Program Fails

The most important question a research program can answer is not "does it work?" but "what do we learn either way?" A program that only produces knowledge when it succeeds is advocacy, not science. Each phase of this program has defined failure criteria that advance understanding regardless of outcome.

If the high-confidence experiments fail — if RSL3 and SDT show similar sensitivity to hypoxia, or if light and sound attenuate similarly in tissue, or if dual-pathway depletion produces additive rather than synergistic kill — the most fundamental predictions of the model are wrong. The ROS-source asymmetry (endogenous versus exogenous) is smaller than the simulation estimates, or it doesn't exist at the treatment-relevant scale. The field learns that the physical-modality advantage for ferroptosis is purely spatial (depth of penetration), not biochemical. This narrows the claim substantially but does not eliminate it: depth advantage is still real for deep-seated tumors.

If the medium-confidence experiments fail — if the vulnerability window is much shorter than predicted, if DAMP release does not correlate with treatment modality, or if CAF co-culture provides no detectable protection — the directional findings from Phase 1 may still hold, but the quantitative magnitudes are wrong and the model's parameter estimates need substantial revision. The field learns what calibration data is needed for the next generation of ferroptosis models.

If the entire program fails — if every in-vitro experiment shows no modality-dependent differential — the directional findings were computational artifacts produced by estimated parameters that do not reflect real biology. This is not a wasted outcome. It defines the limits of computational hypothesis generation in this parameter regime: models with estimated parameters can identify directions worth exploring, but they cannot reliably distinguish real biological patterns from parameter-driven artifacts without experimental calibration. Future modeling efforts in the ferroptosis space would know to calibrate against experimental data before making predictions, not after. That methodological lesson has value beyond this specific hypothesis.

Chapter 10

The Broader Landscape

10.1 Conclusion

This article began as a modality-centered question and ends in a broader place.

What the corpus analysis shows. Cancer-therapy literatures need to be compared on resistant-state relevance, evidence maturity, delivery constraints, and known analytic artifacts before any one mechanism is over-privileged. Our evidence tagger has 96% precision but only 55% recall — meaning tagged evidence is almost always real, but absence claims require qualification. The tissue-of-origin layer reveals that 74% of physical-modality research is concentrated in epithelial tumors, with hematologic and mesothelial contexts nearly absent. The designed-combination classifier shows that only 19.7% of multi-mechanism papers represent actual combination studies — the rest are reviews or incidental co-mentions.

What the simulations show. The mechanistic simulations produce three findings that, if validated experimentally, would have translational implications:

1. *Drug combination synergy through dual-pathway depletion.* RSL3 (GPX4 inhibitor) + FSP1 inhibitor produces 1.99× Bliss synergy because depleting both parallel repair pathways drops total antioxidant defense below the autocatalytic LP propagation threshold. This suggests combination strategies targeting GPX4 AND FSP1 simultaneously, rather than either alone.

2. *The tumor microenvironment selectively favors physical over pharmacologic ferroptosis inducers — through three resistance mechanisms and one amplification effect.* (a) Hypoxia: RSL3 kill collapses from 3.7% to 0.1% under oxygen gradients; SDT maintains 87.8%. (b) Stromal shielding: CAF-mediated GSH/MUFA supply halves RSL3's peripheral kill rate (3.0% to 1.5%); SDT barely affected (96.1% to 91.2%). (c) Acidic pH: ion trapping halves RSL3 ferroptosis kills (163 to 77, 53% reduction) as drug bioavailability drops 36% at pH 6.5; SDT gains only 0.8% (iron boost negligible vs exogenous ROS). These three resistance mechanisms share the same structural asymmetry: pharmacologic

inducers depend on slow endogenous processes that the TME disrupts, while physical modalities deliver exogenous energy that bypasses these vulnerabilities. (d) Immune coupling is a downstream amplification that also favors physical modalities: SDT generates 104× more immune kills than RSL3 (521 vs 5), amplified 2.8× by anti-PD-1 (to 1,477).

3. *Tissue-specific drug penetration creates a substantial in-vitro-to-in-vivo translation gap.* Even at the blood vessel wall, tissue-specific vascular permeability reduces RSL3-like drug kill from 40% (2D culture) to 12.1% (well-vascularized epithelial) to 2.6% (poorly-vascularized pancreatic) to 1.8% (CNS/BBB). The gap is even larger once the penetration gradient into the tissue is considered.

What remains conditional. Within the broader frame, the hypothesis that physical ROS-generating modalities may be useful for locally targetable, ferroptosis-prone residual states survives not because we've proven it, but because the simulation identifies specific conditions under which it holds (hypoxic cores, deep tissue, limited drug penetration) and specific conditions under which it doesn't (hematologic malignancies, tumors without ferroptosis-prone states, settings where ICD-to-immunity chain has historically underperformed). PDT is the established precedent, SDT is the depth-extended variant, and all are constrained by nanosensitizer dependence, uncertain in vivo ferroptosis durability, and the historical failure rate of preclinical immune logic in trials.

The deeper lesson is methodological. Gap claims can be artifacts. Mechanism counts are not maturity. Patient-study signal is not phase evidence. A compelling literature bridge is still only the start of a research program. But the simulation work adds something that literature analysis alone cannot: it identifies which biological conditions would need to be true for the hypothesis to work, and which would falsify it. The RSL3+FSP1i synergy, the hypoxia differential, and the tissue penetration gap are all testable in the lab.

Openness and reproducibility. All code, data, and analysis pipelines are openly available under the MIT license. The ferroptosis-core simulation engine is published as a standalone Rust library with Python bindings, so any researcher can reproduce, challenge, or extend the computational work. We believe this kind of radical openness — where the tools are as available as the conclusions — is essential for research that aims to help people rather than to be right about one hypothesis.

The right next step is therefore not to defend one favored modality, but to run the comparative experiments and cross-lane evaluations that could falsify it quickly.

10.2 Where the Field Is Going

The cancer research landscape in 2026 is defined by several structural trends that provide context for where this work sits and where it might become irrelevant.

Immunotherapy remains the dominant paradigm, but resistance is the

dominant problem. Checkpoint inhibitors have transformed outcomes for some cancers — melanoma, non-small-cell lung cancer, renal cell carcinoma — and their use continues to expand. But the core challenge has shifted from "does immunotherapy work?" to "why doesn't it work for most patients?" Primary resistance (tumors that never respond), acquired resistance (tumors that respond and then escape), and the immunosuppressive tumor microenvironment are now the central obstacles. The same TME features modeled in Chapter 7 — hypoxia, stromal shielding, acidic pH — are recognized as barriers to immunotherapy effectiveness, not just to ferroptosis induction.

Engineered cell therapies face the solid-tumor wall. CAR-T cells have been transformative in B-cell leukemias and lymphomas, but extending their success to solid tumors remains the field's greatest challenge. The tumor microenvironment suppresses, exhausts, and excludes T cells attempting to infiltrate solid tumors. Every advance in CAR-T for solid tumors is, at root, an advance in overcoming TME suppression — the same biological landscape this project models from a different angle.

Radioligand therapy is the clearest clinically advancing competitor to any spatially selective hypothesis. Lutetium-177 PSMA-617 (Pluvicto) is FDA-approved for metastatic castration-resistant prostate cancer. Lutetium-177 dotatate (Lutathera) is approved for gastroenteropancreatic neuroendocrine tumors. The theranostic pipeline — using imaging to identify targetable tumors and then treating with the same targeting molecule carrying a radioactive payload — is more clinically mature than anything in the ferroptosis-SDT space. For tumors with targetable receptors, radioligand may already offer what the physical-ROS hypothesis proposes — spatially selective killing with systemic delivery.

AI and computational biology are accelerating hypothesis generation. Machine learning is transforming drug target identification, compound screening, and biomarker discovery. This project is one instance of a broader trend: using computational tools to map research landscapes and identify gaps that individual researchers miss. The specific tools (Monte Carlo biochemical simulation, cross-literature corpus analysis) are distinctive, but the underlying approach — computational hypothesis generation followed by experimental validation — is becoming standard practice.

mRNA cancer vaccines are advancing toward personalized treatment. Individualized neoantigen vaccines that train the immune system to recognize mutations specific to each patient's tumor are in late-stage clinical trials. This approach addresses the immune-recognition problem directly rather than relying on immunogenic cell death to trigger an immune response. If personalized mRNA vaccines prove effective at eliminating residual disease, the ICD component of the physical-ROS hypothesis becomes less important — though the direct cytotoxic advantage of physical modalities in hypoxic tumors would remain.

10.3 Where This Work Fits — and Where It Doesn't

This project contributes a computational framework for comparing therapeutic approaches across the boundaries that normally separate research communities. Its value lies in the questions it raises and the comparisons it surfaces, not in clinical claims.

Where it fits. The hypothesis generation is most relevant for locally targetable, deep-seated solid tumors with evidence of ferroptosis-prone residual states — specifically, tumors where pharmacologic drug delivery faces TME barriers (hypoxia, stromal shielding, acidic pH) that the simulation predicts physical modalities can bypass. Pancreatic, hepatic, and pelvic tumors in immunocompetent patients during a post-treatment vulnerability window are the narrowest and most defensible application context.

Where it does not fit. Hematologic malignancies (no spatial targeting needed). Diffuse metastatic disease (no localizable target for focused ultrasound or light). Tumors without ferroptosis-prone residual states (no vulnerability to exploit). Immunocompromised patients (ICD-to-immunity chain cannot operate). Settings where the vulnerability window has closed (persister cells have already restored defenses). This is not a general theory of cancer treatment. It is a conditional prediction about a specific clinical context.

The honest framing. This work identifies questions worth asking, not answers worth implementing. Every simulation finding is a hypothesis, not a demonstrated clinical advantage. The experimental program in Chapter 9 is designed to determine which — if any — of these hypotheses survive contact with real biology. If none do, the contribution is the computational framework and the cross-literature methodology, which remain useful for other hypotheses and other research communities.

10.4 Competing Hypotheses Worth Taking Seriously

Intellectual honesty requires acknowledging that the ferroptosis-persister-physical-ROS hypothesis could be rendered irrelevant by developments in adjacent fields, independent of whether its specific predictions hold.

The ferroptosis-persister connection might be a niche phenomenon. The evidence that persister cells are ferroptosis-sensitive comes from specific model systems (primarily EGFR-mutant lung cancer persisters). Whether this sensitivity generalizes across cancer types, treatment histories, and genetic backgrounds is unknown. If it doesn't generalize, the hypothesis applies to a narrow population rather than the broad residual-disease problem described in Chapter 1.

Radioligand therapy might render the spatial-selectivity argument unnecessary. For tumors with targetable surface receptors, radioligand delivers spatially selective killing through systemic injection — no focused ultrasound or light source required. If the receptor-targeting approach expands to cover more tumor types, the physical-

modality advantage narrows to the subset of tumors without targetable receptors.

Immune checkpoint combinations might solve residual disease independently. If next-generation checkpoint combinations (anti-PD-1 + anti-CTLA-4 + anti-LAG-3, or checkpoint plus costimulatory agonists) prove effective at eliminating minimal residual disease, the ferroptosis-ICD pathway becomes one of many ways to achieve immune clearance rather than a uniquely valuable approach.

The nanoparticle engineering confound might mean there is no SDT-as-modality advantage. As Chapter 8.3 noted, most SDT-ferroptosis data comes from engineered nanoparticles designed to deplete GSH and release iron. If plain SDT with conventional sonosensitizers does not trigger ferroptosis, the entire hypothesis is about nanoparticle design, not about ultrasound as a therapeutic modality.

The field may move past the ferroptosis-persister hypothesis entirely. If it does, this project's lasting contribution would be the computational methodology — the ferroptosis-core simulation engine, the cross-literature analysis pipeline, and the framework for comparing therapeutic approaches on shared axes. Those tools are useful regardless of whether the specific hypothesis they were built to test turns out to be correct.

Chapter 11

How To Contribute

This project is built to be extended, challenged, and improved. Everything — the code, the data, the analysis pipeline, and this manuscript — is publicly available under the MIT license. This chapter describes specific ways to engage, organized by background.

11.1 For Researchers

The ferroptosis-core simulation engine is a standalone Rust library designed to be used independently of this manuscript. It models the ferroptosis cascade (ROS generation, GSH depletion, GPX4/FSP1 repair, lipid peroxidation, and cell death) with configurable parameters and stochastic variation.

The simplest way to engage is to challenge the predictions. Every simulation finding in Chapters 6 and 7 depends on specific parameter values documented in the parameter provenance file (see ‘simulations/calibration/parameter_provenance.md‘). Changing a parameter and rerunning the simulation tests whether the directional result survives. If it doesn't, you've identified a load-bearing assumption — which is exactly what the experimental program in Chapter 9 aims to do with wet-lab experiments. The Python bindings (‘pip install ferroptosis-core‘) make this accessible without learning Rust: import the library, call ‘sim_cell‘ or ‘sim_batch‘ with modified parameters, and compare against the published results.

If you have experimental data that could calibrate model parameters — measured GPX4 recovery kinetics after drug withdrawal, quantified CAF supply rates of GSH or MUFA, or RSL3 intracellular concentration at different pH values — those measurements would directly improve the model's parameter estimates and change the confidence tiers in Chapter 9. The project welcomes data contributions as GitHub issues linking to published or preprint data.

If you find articles missing from the local corpus that would materially change the analysis — particularly landmark papers that the corpus-gaps document (‘analysis/landmark-

corpus-gaps.md‘) already identifies — submitting a pull request or issue with the PMID helps close known coverage gaps.

11.2 For Engineers and Developers

The simulation architecture is modular by design. The ferroptosis-core library (see ‘simulations/ferroptosis-core/README.md‘) exposes eight modules: ‘cell‘ (types and phenotypes), ‘params‘ (rate constants), ‘biochem‘ (core engine), ‘stats‘ (confidence intervals and parallel Monte Carlo), ‘physics‘ (depth-dependent energy), ‘grid‘ (2D tumor architecture), ‘immune‘ (ICD/DAMP cascade), and ‘io‘ (JSON/CSV output). Each module can be extended independently.

Adding new TME features. The sim-tme binary demonstrates how to compose ferroptosis-core with spatial features (O2 gradients, pH fields, stromal layers, immune coupling). Adding a new feature follows the same pattern: define the spatial field, modify cell parameters per-step based on the cell's position, and run the existing biochemistry engine. The adaptive-evolution model identified as missing in Chapter 8.4 — where surviving cells mutate and evolve resistance across treatment cycles — would be the most impactful extension.

Python bindings. The ferroptosis-python package (see ‘simulations/ferroptosis-python/README.md‘) wraps the Rust engine via PyO3. Install with ‘pip install ferroptosis-core‘ (no Rust toolchain required). The API exposes ‘default_params()‘, ‘invivo_params()‘, ‘sim_cell()‘, and ‘sim_batch()‘ with full parameter override support. Performance is native Rust speed with the GIL released during computation.

C FFI for PhysiCell integration. The ferroptosis-ffi crate (see ‘simulations/ferroptosis-ffi/‘) provides a C-compatible API for embedding the ferroptosis engine in agent-based modeling platforms like PhysiCell. This enables multi-scale simulations combining ferroptosis biochemistry with cell migration, proliferation, and tissue-level dynamics that the current model does not capture.

11.3 For Students

This project is designed to be a starting point, not a finished product. Several of the limitations identified in Chapter 8.4 translate directly into thesis-level research projects.

Computational biology thesis ideas:

Calibrate the vulnerability window model against experimental data (extends Chapter 6.2). The current model uses literature-estimated recovery half-lives for GPX4, FSP1, NRF2, and GSH. A project that measures these recovery kinetics in a specific persister cell model and re-parameterizes the simulation would convert a medium-confidence prediction

into a high-confidence one — or reveal that the model's assumptions are wrong. Skills needed: cell culture, Western blot/qPCR for protein quantification, basic Rust or Python for model modification.

Build the systemic immune model that Chapter 8.4 identifies as missing (extends Chapter 7.2). The current immune layer models resident T cell activation over 0-48 hours. A systemic model incorporating circulating immune cell recruitment, regulatory T cell suppression, and adaptive memory formation would resolve the "amplification vs dampening" uncertainty identified in Chapter 8.4. Skills needed: computational modeling, immunology background, Rust programming.

Add adaptive evolution to the simulation (addresses Chapter 8.4's structural limitation). The current model simulates one treatment cycle. A population-dynamics extension that models mutation, clonal selection, and resistance evolution across multiple cycles would answer the durability question the current model cannot address. Skills needed: evolutionary dynamics, agent-based modeling, Rust or Python.

11.4 For Clinicians

This project does not make treatment recommendations. Nothing in this book should be interpreted as clinical guidance. The simulation predictions are hypothesis-generating tools, not validated clinical evidence.

What clinicians can watch for: any clinical trial that tests ferroptosis-inducing agents in hypoxic solid tumors would provide real-world evidence for or against the predictions in Chapter 7. Any trial combining SDT with immunotherapy would test the immune-coupling prediction from Section 7.2. Any trial measuring ferroptosis markers as secondary endpoints after physical-modality treatment would provide calibration data for the model.

How to interpret the simulation predictions: directional findings (physical modalities bypass TME resistance better than drugs) are the claims most worth following — they are robust to parameter variation and represent the model's strongest predictions. Quantitative values (the specific percentages, ratios, and synergy scores) should be treated as order-of-magnitude estimates, not precise forecasts. The confidence tiers in Chapter 9 provide a guide to which predictions rest on the strongest evidence.

The contribution clinicians can make is judgment. Computational biologists can identify patterns in data; clinicians can judge whether those patterns matter in the context of real patient care, real treatment logistics, and real clinical endpoints. If the experimental program in Chapter 9 produces results, clinician input on study design for the in-vivo integration experiment (Section 9.5) would be valuable — particularly on endpoint selection, comparator arm design, and the clinical context most likely to test the hypothesis fairly.

Part V

References and Tools

[Appendices A-C to be added in #104]

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